BACnet/IP Simulation System

BOSS: User Manual

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Distribution List

Name	Description
Softdel and Softdel's Client.	Softdel and Softdel's Client teams using Softdel's BOSS simulator.

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1 Introduction

BACnet (Building Automation and Control Network) is a communication protocol that defines services used to communicate between building automation control systems and the end-devices.

Building automation solution providers and smart building OEMs are constrained by the lack of availability of comprehensive tools to establish BACnet communication within their product offerings. Most available tools have not kept up with the continuously evolving BACnet standard. In addition, BACnet application development requires physical reference to BACnet devices to be able to test applications in real-time. Most often, interoperability and testing only occur in the BACnet test lab at the time of certification or directly during field deployment. Because of this environment, BACnet building automation providers and smart building OEMs spend a significant amount of time and expense in troubleshooting BACnet integration into their devices and applications for market readiness.

Softdel has a Product which is a PC based tool called as **BOSS – BACnet/IP Simulation System**.

The BACnet device simulator (BOSS) toolset aims to bring a comprehensive ability to effectively simulate BACnet functionalities and interoperability by simultaneous simulation of multiple virtual device configurations. BOSS's easy to use interface allows developers and system integrators to configure, monitor and issue service commands to accessible physical and virtual devices on the BACnet network. In addition to this BOSS also provides BBMD, FD and Transaction-Log functionalities.

Advantages of using BOSS are as follows:

- Helps development and testing team to understand various aspects of BACnet standard.
- Users can create multiple virtual devices having different configurations, on a single desktop
- Reduction in development cost
- Faster product life cycle execution

Features of BOSS are as follows:

- User friendly graphical user interface
- Supports addition of multiple virtual BACnet/IP devices
- Each virtual device can work independently over the BACnet network
- Supports A side and B side services
- Supports various objects and properties
- Supports object simulation
- Supports different character sets
- Supports segmentation
- Supports BBMD
- Transaction log
- Monitoring of objects properties
- Project and virtual device data can be saved for later use.

1.1 Purpose and Scope

Purpose of this user manual is to provide information on usage of Softdel's BACnet/IP Simulation System i.e. BOSS. This manual provides description of the system functions and capabilities, contingencies and alternate modes of operation, and step-by-step procedures for system access and use.

Scope of this manual is limited to usage of BOSS tool only. Information related to other things like BACnet network setup, administrative setup on Windows, firewall setup, etc. is not is the scope of this document.

1.2 Abbreviations / Terms

Abbreviation	Description
APDU	Application Protocol Data Unit
B/IP or B-IP	BACnet over Internet Protocol
BBMD	BACnet Broadcast Management Device
BDT	Broadcast Distribution Table
BOSS	BACnet/IP Simulation System
CLI	Command Line Interface
COV	Change Of Value
DCC	Device Communication Control
FD	Foreign Device
FDT	Foreign Device Table
IP	Internet Protocol
MAC	Media Access Control
OS	Operating System
RAM	Random Access Memory
RD	Reinitialize Device
UDP	User Datagram Protocol

Table 1: Abbreviations

2 Installation

This chapter provides details of BOSS installation procedure and BOSS requirement. BOSS is distributed in the form of an executable Windows installer file.

BOSS has some hardware and software requirements; these are mentioned below.

2.1 Hardware Requirements

Hardware requirements are as mentioned below:

- Processor speed (minimum) – 1.7 GHz
- RAM size (minimum) – 4 GB
- Display resolution (recommended) – 1024 * 768 and above.
- One active NIC (network interface card)

2.2 Operating System Requirements

Operating system requirements are as mentioned below:

- Windows 10 and above

2.3 Supported Architectures

BOSS supports following hardware architectures:

- x86
- x64 (WoW64)

2.4 Pre-Requisite

BOSS installation requires following Windows components to be pre-installed on the computer.

- Windows Installer 4.5 or later
- Microsoft .NET Framework 4.8.1 or later
- VC++ Redistributable 2017 x86

2.5 Installation Guide

This section describes the steps to install the BOSS Simulator on the user's system.

Steps to install are as follows:

Note: the user may need administrative privileges on Windows OS, to install and run BOSS tool.

Step 1:

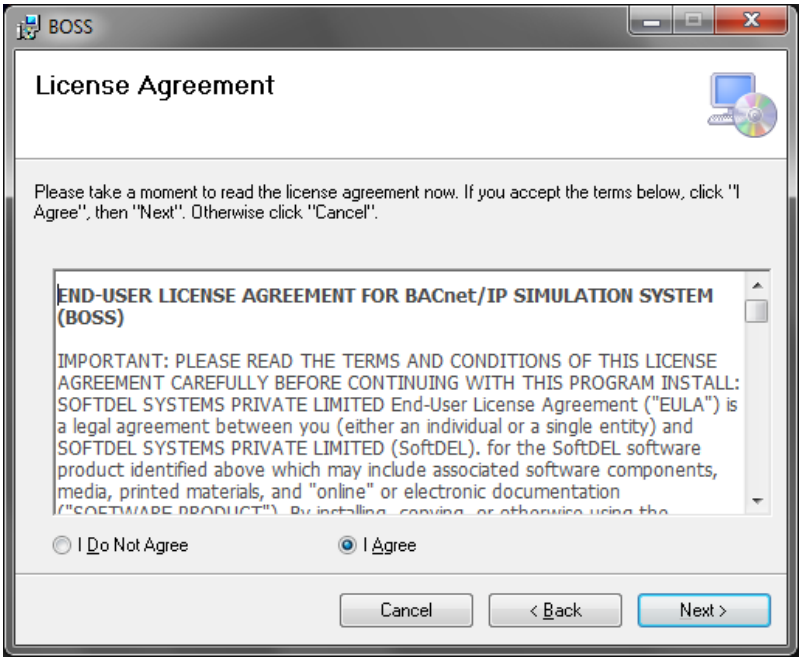
Run **setup.exe** executable file.

Below screen will appear on running the executable file.

Click on **Next** button to proceed.

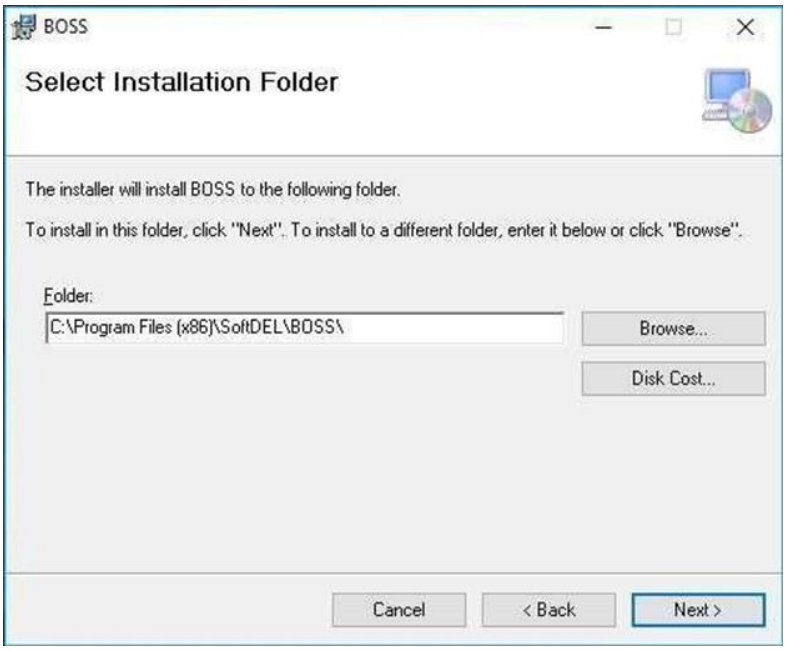
Step 2:

Select **I Agree** radio button and then click on **Next** button.



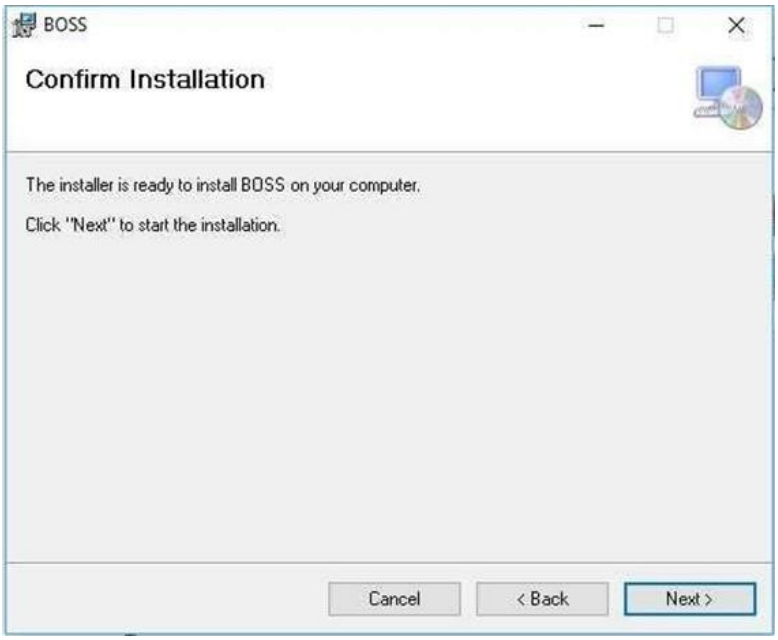
Step 3:

Select the installation folder, default path for installation is already set, user can change this if required. If user wants to change the path, it can be done by clicking **Browse** button on the screen. Once the destination path is set, click on the **Next** button to start installation.



Step 4:

Click on **Next** button to proceed for installation.

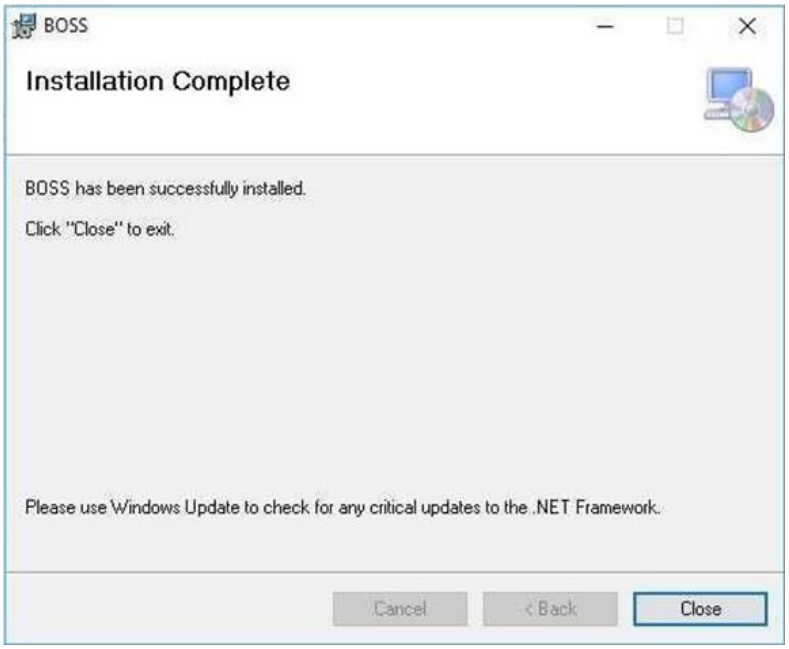


Step 5:

Following screen appears once the installation is completed.

Click **Close** button.

Shortcut to BOSS will be created on the desktop and in Windows Start Menu.



3 Registration

After installing BOSS, the user is required to register with Softdel Systems Pvt. Ltd. Upon launching BOSS for the first time, the following screen is displayed.

The registration process proceeds as follows:

Step 1:

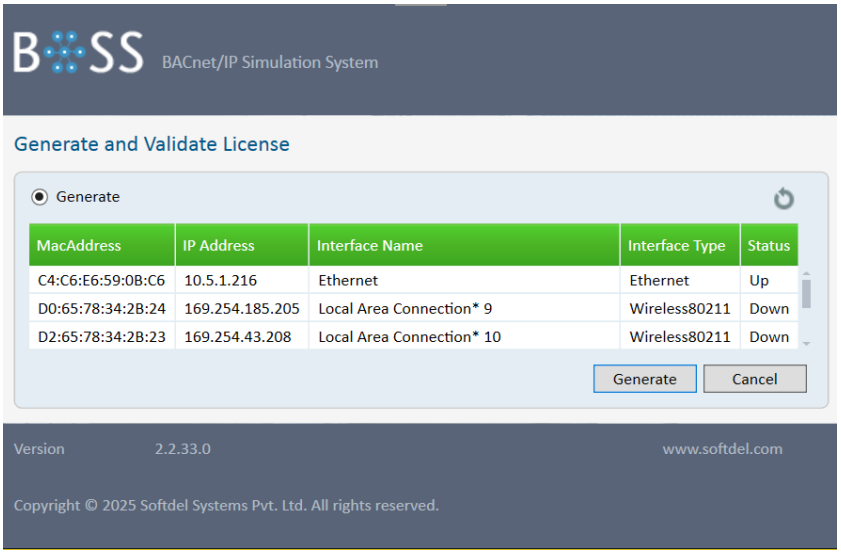
Click the **Register** button to begin the BOSS registration process.

Step 2:

After clicking the **Register** button, the user is navigated to the **Network Interface** page. Here, the user must select a **network interface** that meets the following criteria:

- The interface must be **active (up)**.
- The MAC address must belong to the **user's machine** only.

After selecting the correct network interface, click the **Generate** button. The application will open a **Save File** dialog prompting the user to save a .bnt file.



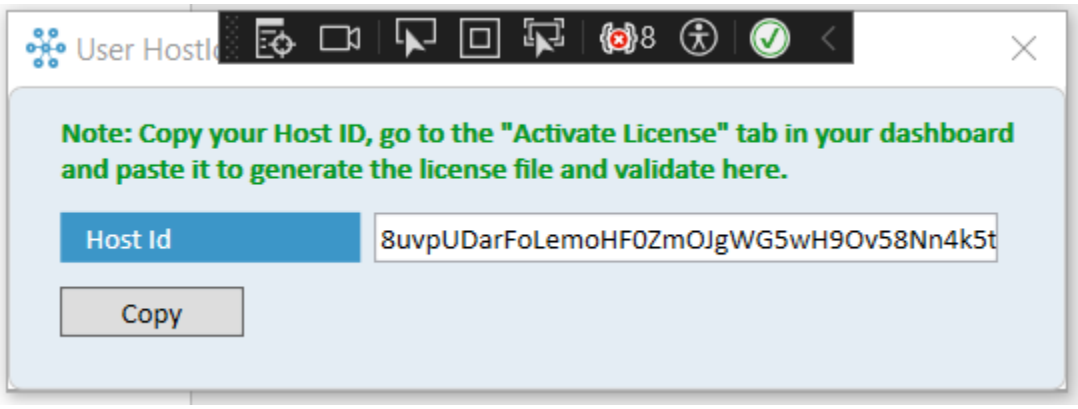
Step 3:

Email Host Id to **info@softdel.com** with the following details included in the message body:

1. Full name of the user
2. Email address
3. Company name

Use the subject line: **"BOSS License Request"**

For any clarifications or support, feel free to write to info@softdel.com

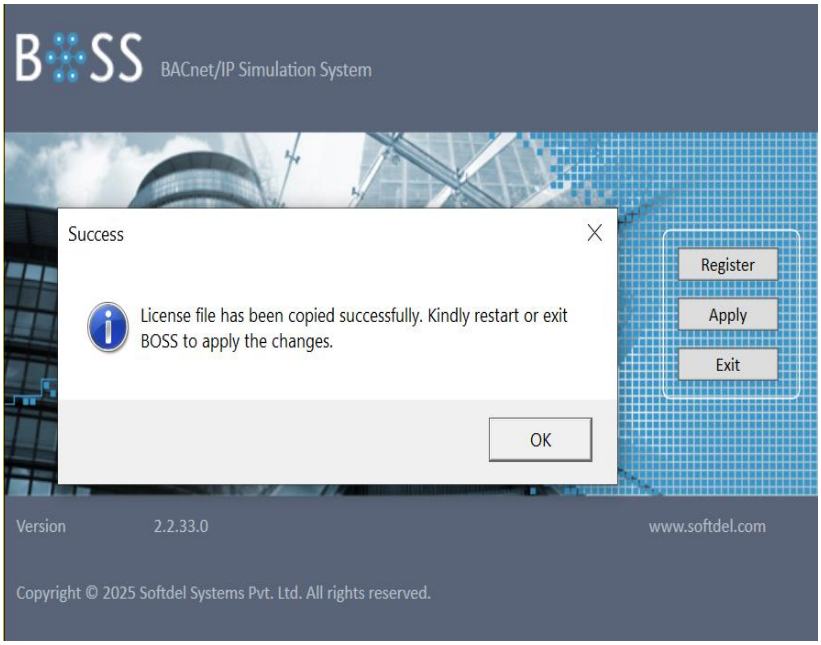


Step 4:

After receiving the Host Id, **Softdel Systems** will provide the corresponding license file to the user.

Step 5:

To validate and apply for the license, click the **Apply** button and select the .lic file provided by Softdel. Once the license is successfully validated and copied to the required path, the user will receive confirmation through a **notification pop-up**.



Step :6

Upon successful validation of the license, the following screen will be displayed.

If the screen from **Step 1** continues to appear even after applying the license, please contact the **BOSS Support Team** at boss-support@softdel.com or info@softdel.com for assistance.

4 Feature List

Various features provided in the BOSS, are mentioned in this chapter.

Sr. No.	Feature Name
1	Device Configuration
2	Addition/Deletion/Replication of Virtual Devices
3	Addition/Deletion of Objects
4	Simulation
5	Segmentation support
6	Character set support
7	Device discovery
8	Object discovery
9	Property discovery
10	Property modification
11	Virtual Routing
12	A-side and B-side services
13	BBMD, Non-BBMD and FD
14	COV
15	Backup and Restore
16	Alarm and Event: Intrinsic Reporting
17	Alarm and Event: Algorithmic Reporting
18	Trending
19	Scheduling
20	Project open/save/create
21	Multiple BOSS instance creation on single computer
22	Command line interface
23	Scheduling using CSV files
24	Project file replication
25	BACnet Profile wise virtual device(s) creation using predefined CSV based library
26	Device creation using user defined CSV file

Table 2: BOSS Features

4.1 BACnet Functionalities

4.1.1 Object Types Supported

BACnet objects supported in BOSS are given in below table. These objects can be created in virtual devices of BOSS.

No.	Object Type
1	Device
2	Network Port
3	Analog Input
4	Analog Output
5	Analog Value
6	Binary Input
7	Binary Output
8	Binary Value
9	Multi State Input
10	Multi State Output
11	Multi State Value
12	Notification Class
13	Event Enrollment
14	Calendar
15	Schedule
16	File
17	Trend Log
18	Loop
19	Accumulator

Table 3: BOSS Supported Objects

Note that BOSS will be able to read and write properties of other object types present in network.

4.1.2 Services Supported

BACnet services supported in BOSS are given below table.

No.	Service Classification	Service Description	Initiate	Execute
1	**Alarm & Event**	Acknowledge Alarm	×	✓
2		Confirmed COV Notification	×	✓
3		Unconfirmed COV Notification	×	✓
4		Confirmed Event Notification	×	×
5		Unconfirmed Event Notification	×	×

No.	Service Classification	Service Description	Initiate	Execute
6		Get Alarm Summary	✓	×
7		Get Enrollment Summary	×	×
8		Get Event Information	✓	✓
9		Subscribe COV	✓	✓
10		Subscribe COV Property	×	×
11	**File Access**	Atomic Read File	×	✓
12		Atomic Write File	×	✓
13	**Object Access**	Add List Element	×	✓
14		Remove List Element	×	✓
15		Create Object	×	✓
16		Delete Object	×	✓
17		Read Property	✓	✓
18		Read Property Multiple	✓	✓
19		Read Range	✓	✓
20		Write Property	✓	✓
21		Write Property Multiple	×	✓
22	**Remote Device Management**	Device Communication Control	✓	✓
23		Reinitialize Device	✓	✓
24		Time Synchronization	✓	✓
25		UTC Time Synchronization	✓	✓
26		Who-Is	✓	✓
27		I-Am	×	✓
28		Who-Has	✓	✓
29		I-Have	×	✓
30	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
31	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 4: BOSS Supported Services

5 Getting Started

BOSS provides various functionalities. Procedures to use these functionalities are mentioned in this chapter.

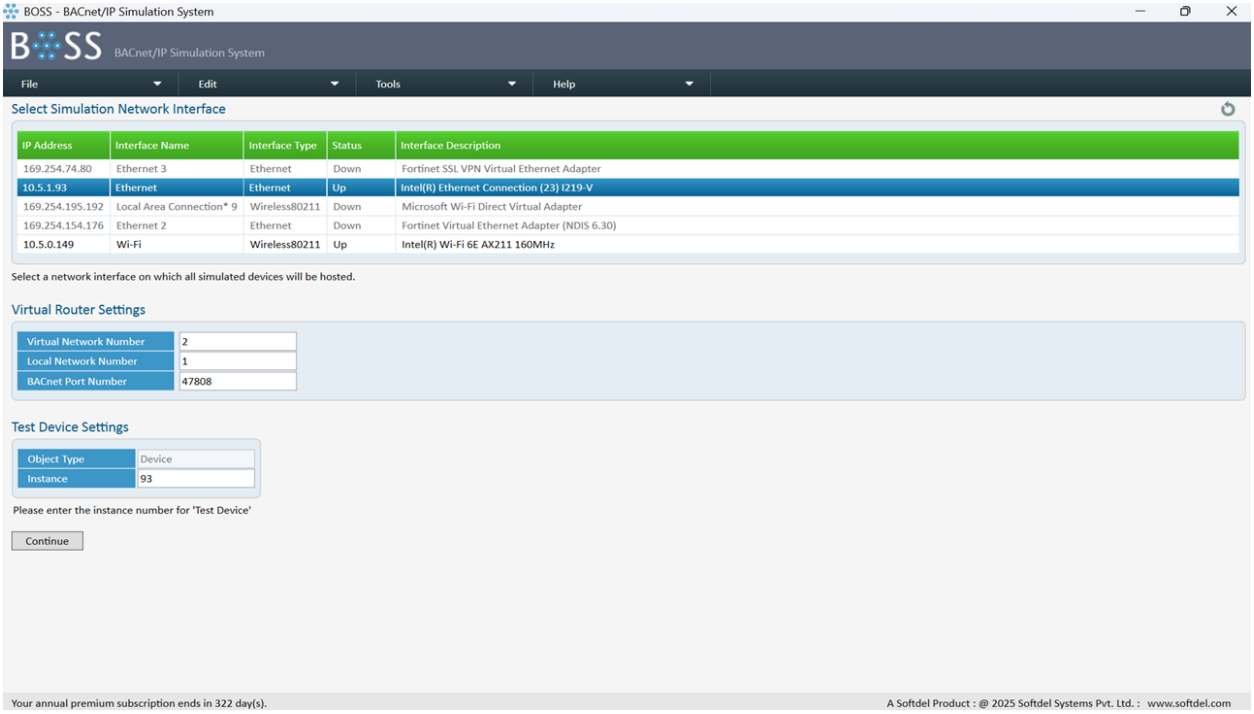
5.1 Starting BACnet/IP Simulation System

Step 1:

Run BOSS application. Following screen appears. Click on **Start** button.

Step 2:

Captured Interface screen appears and lists down available IP addresses present on the system.



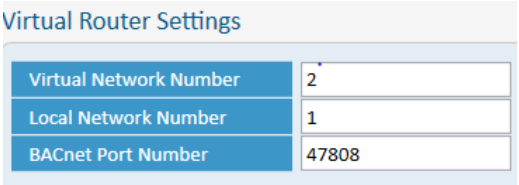
Users need to select an IP address from the list to initiate communication with the BACnet/IP network. BACnet/IP uses unicast and broadcast UDP/IP communication for communicating with other BACnet/IP devices. Hence, the OS Firewall should allow reception and transmission of both unicast and broadcast messages on selected IP-PORT. Kindly refer to Windows help, for configurations related to Windows Firewall.

By default, first row's IP address is selected automatically where status is up.

Refresh Network Interface button on the top right of network interface grid to refresh the network collection and updates the current network status.

Step 3:

Configure **Virtual Router** settings.



Users can edit following items:

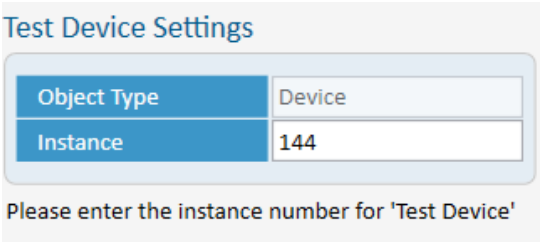
1. Virtual Network Number
   The default Virtual Network number is 2
   Users can configure network number. Allowed range is 2 to 65534.
2. Local Network Number
   The default Local Network Number is 1

Users can configure network number. Allowed range is 1 to 65534.

3. BACnet Port number
   The default BACnet Port number is 47808.
   Users can configure port number. Allowed range is 1 to 65534.

Step 4:

Configure **Test Device** settings.



A device with provided instance number will be created when BOSS is initialized.

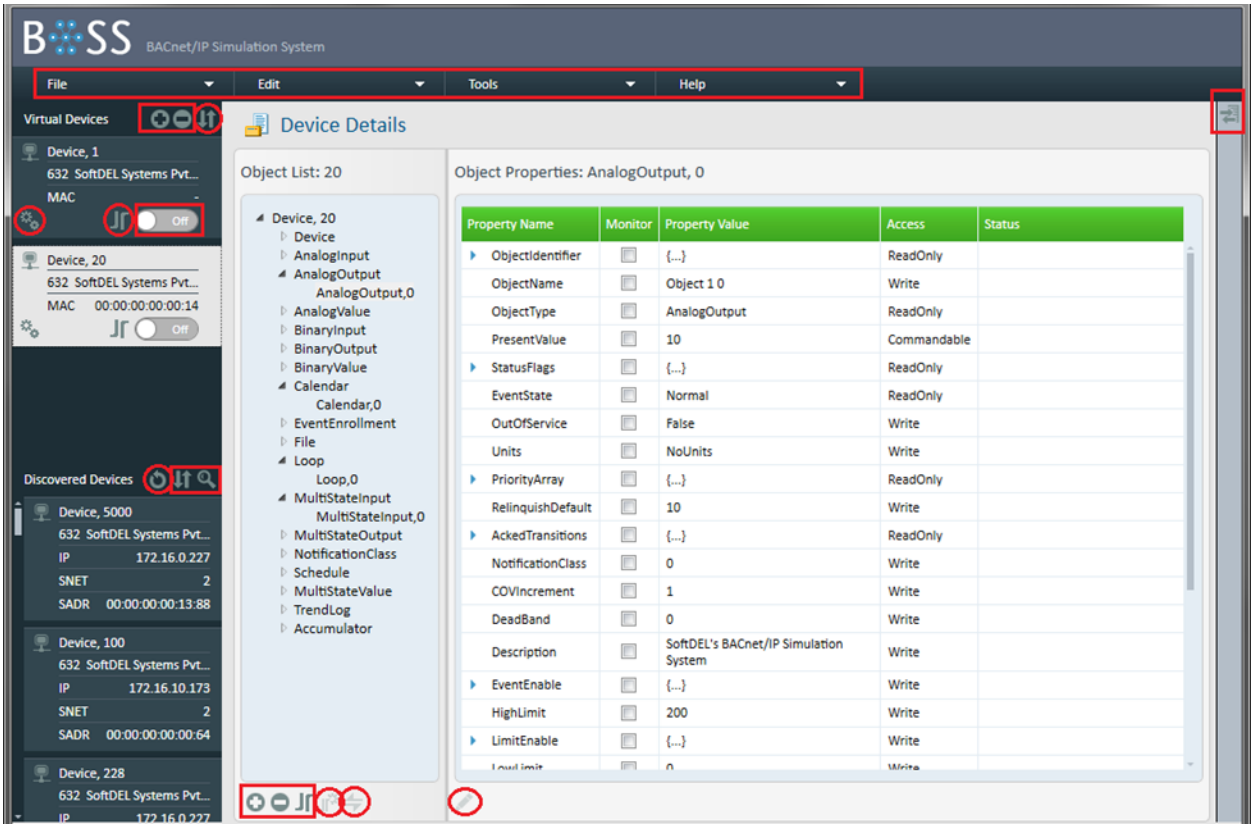
Step 5:

Click **Continue** button to start BACnet communication on selected network interface.

When starts, BOSS will create one device object and will discover accessible devices present in the network.

5.2 Main Window

Main screen of the BOSS appears once user has configured basic settings. Main screen and its details are provided below:



By default, one virtual device is created in the BOSS. The device has instance number configured previously.

Note:

This device is used as a base device to initiate BACnet requests from BOSS. User is allowed to configure few settings of virtual device(s), such as changing instance number, password, segmentation support, etc. Details are provided in 'Device Configuration' chapter.

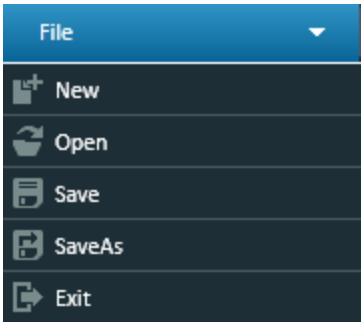
Details of various options present on main window are listed below.

Menu Bar

Menu bar provides various options to access different functionalities available in BOSS.

File Menu

This option provides different options for creating, saving and opening new project.



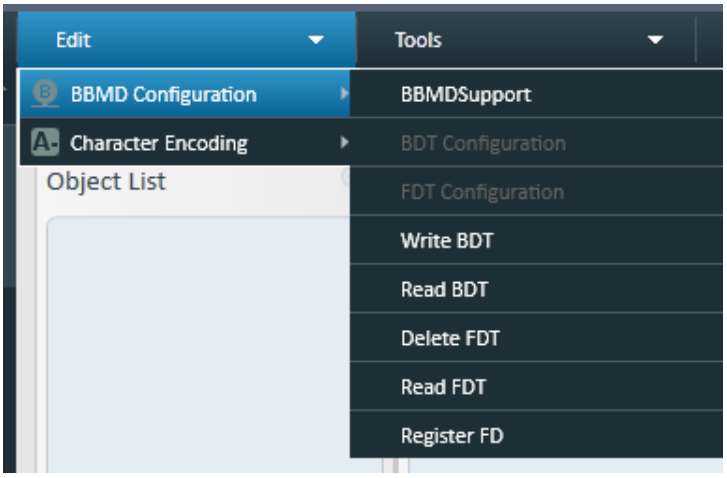
Note:

When using file menu options like 'Save', 'SaveAs' or 'Open, functionalities like simulation, unsolicitedCOV should be OFF. For details of these functionalities, refer 'Device Configuration' chapter.

Edit Menu

Edit menu provides two options.

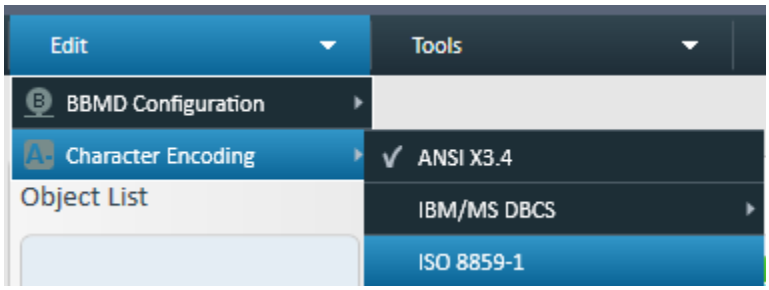
Users can enable or disable BBMD support with the help of **BBMD Configuration** menu.



When user checks BBMD support, BOSS acts like a BBMD device and when unchecked, BOSS acts like a Non-BBMD device, on network.

Register FD functionality is enabled only when BBMD functionality is disabled.

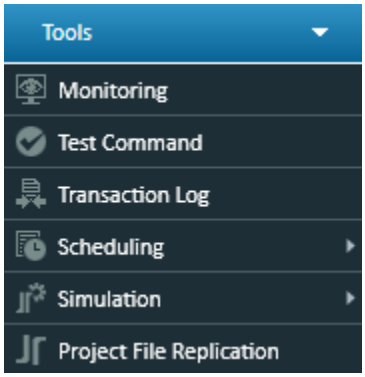
Edit menu also provides an option to select **Character Encoding** format.



Tools

Tools menu provides different options mentioned below:

1. Monitoring
2. Test Command
3. Transaction Log
4. Scheduling
5. Simulation
6. Project File Replication

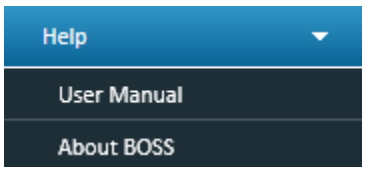


Details of above sub-menus are provided in further chapters within the document.

Help

Help menu provides different options mentioned below:

1. User Manual
2. About BOSS



Note:

Context menu is available for following windowpanes in main window:

'Virtual device list', 'discovered device list', 'object list' and 'object properties'.

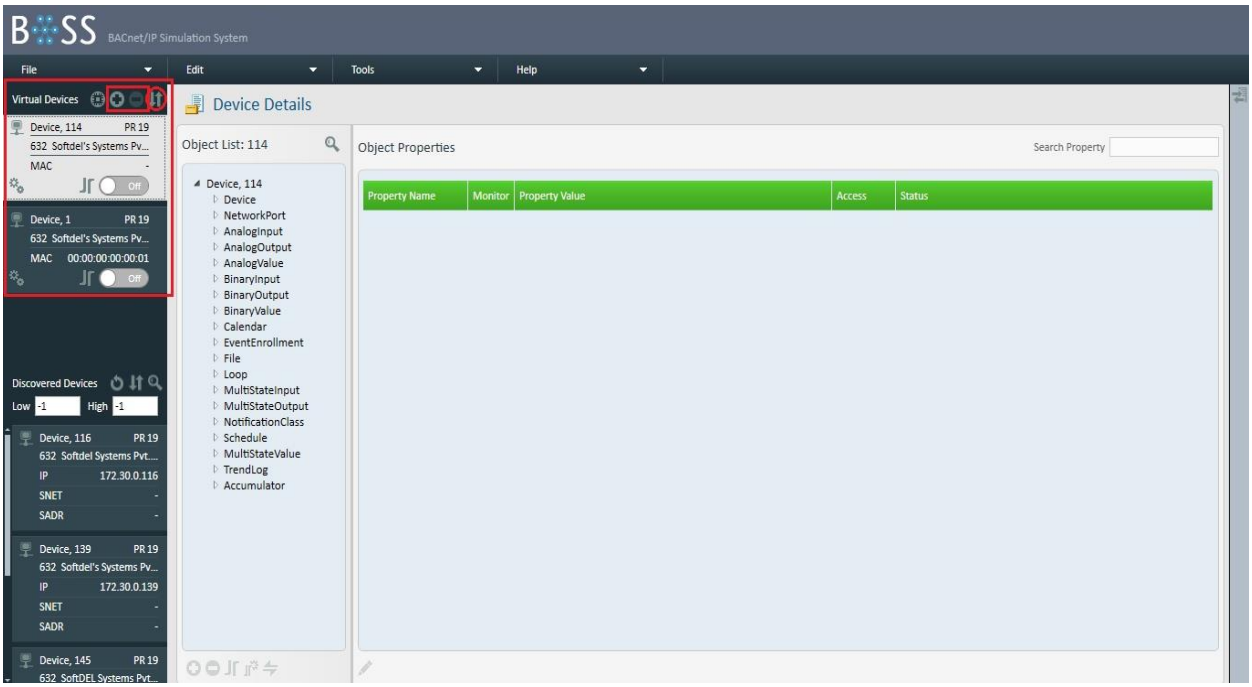
Virtual Device List

Displays all virtual devices created in BOSS.

Users can Add and Delete (except 1st device) virtual devices at runtime. Details of these functionalities are provided in 'Add, Delete and Replicate Virtual Devices' chapter.

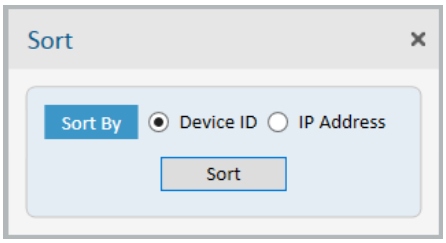
Users can sort virtual device list by device ID, using **Sort Virtual Device** icon. Sorting will be done in ascending order first time and in descending order second time. Pressing sort icon will toggle sorting operation henceforth.

Discovered Device List

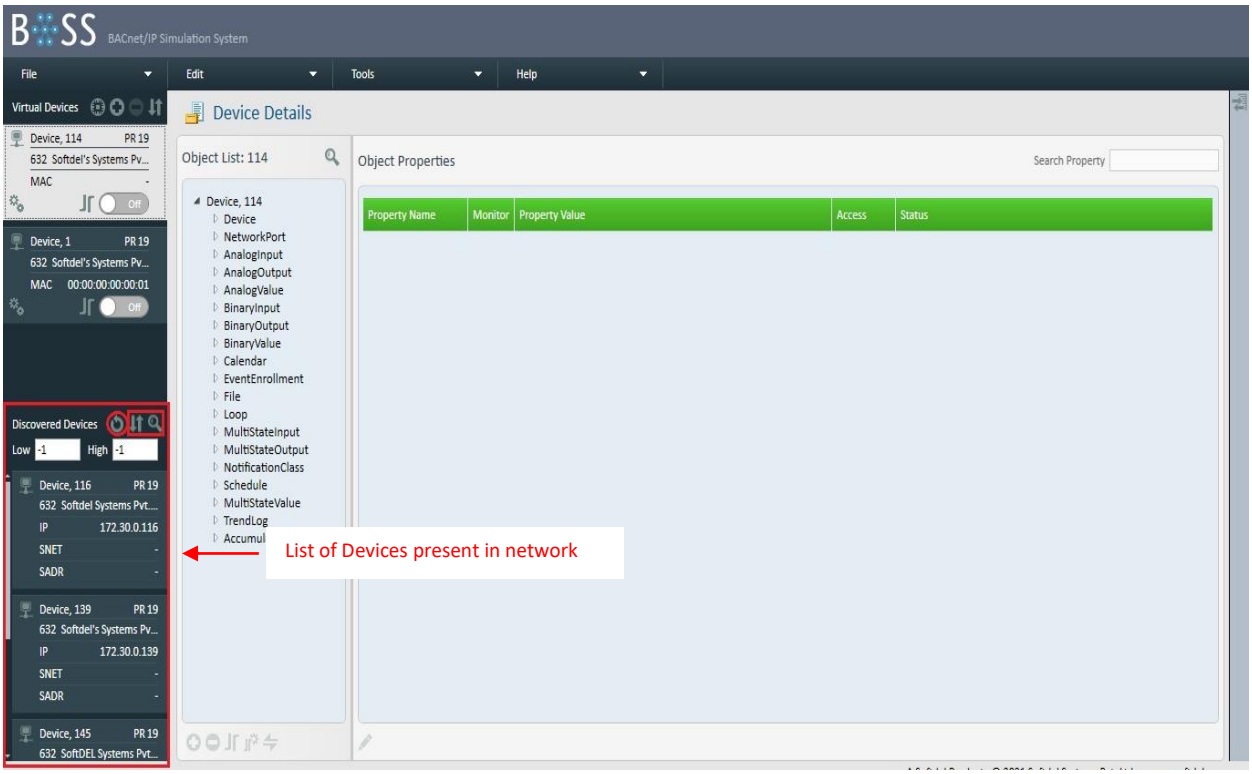


Displays all accessible BACnet devices present on the network. User can discover devices present in network at runtime, using **Load Discovered Device** icon.

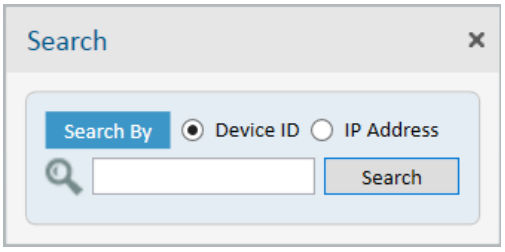
User can sort discovered device list by device ID or by IP address, using **Sort Discovered Device** icon. Following window appears when user clicks on sort icon.



Users can select type of sorting, using respective radio buttons. Sorting will be done in ascending order first time and in descending order second time. Pressing sort button will toggle sorting operation henceforth.



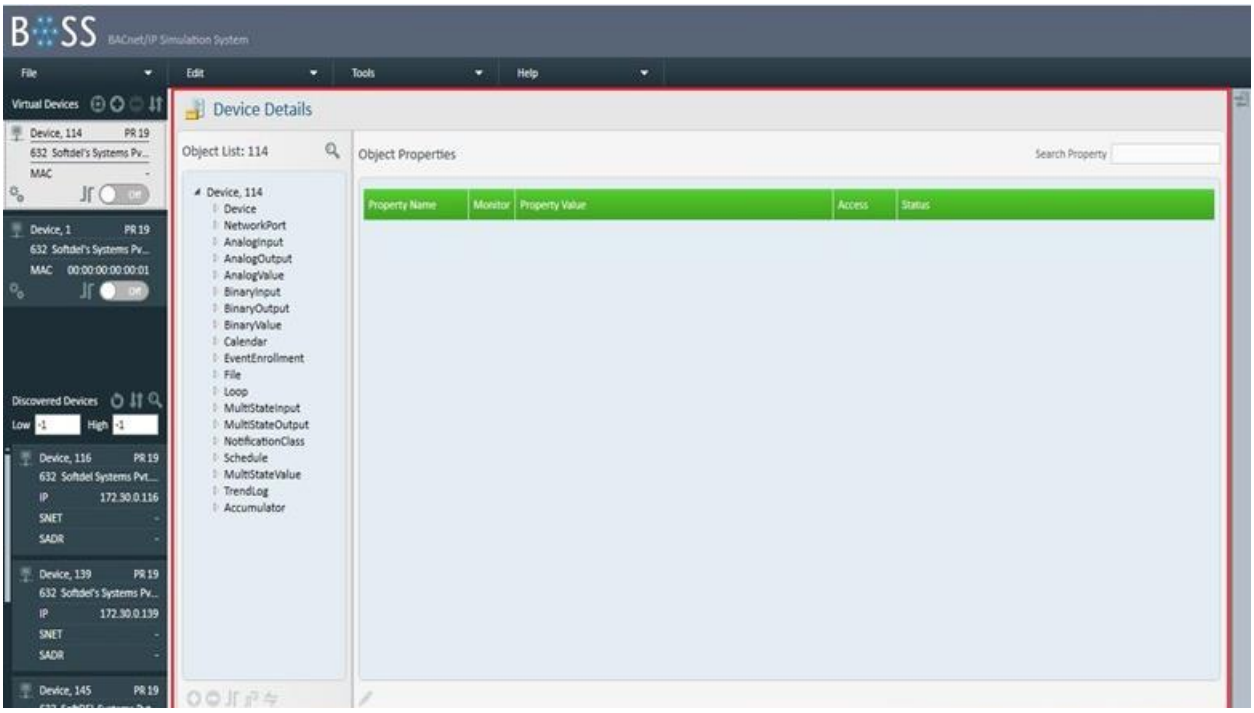
User can search specific discovered device, in discovered device list by device ID or by IP address, using **Search Discovered Device** icon. Following window appears when user clicks on search icon.



User can select type of searching, using respective radio buttons. If searched device is found, it will be displayed at the top of discovered device list. If searched device is not found, error will be displayed.

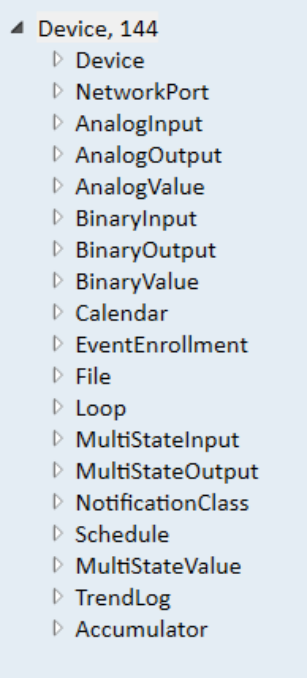
5.3 Device Details Window

The Device Details window displays information of the device, when user selects (clicks on) a device.



Object List Window Pane

This window pane displays all objects of selected device and all instances of each object type.

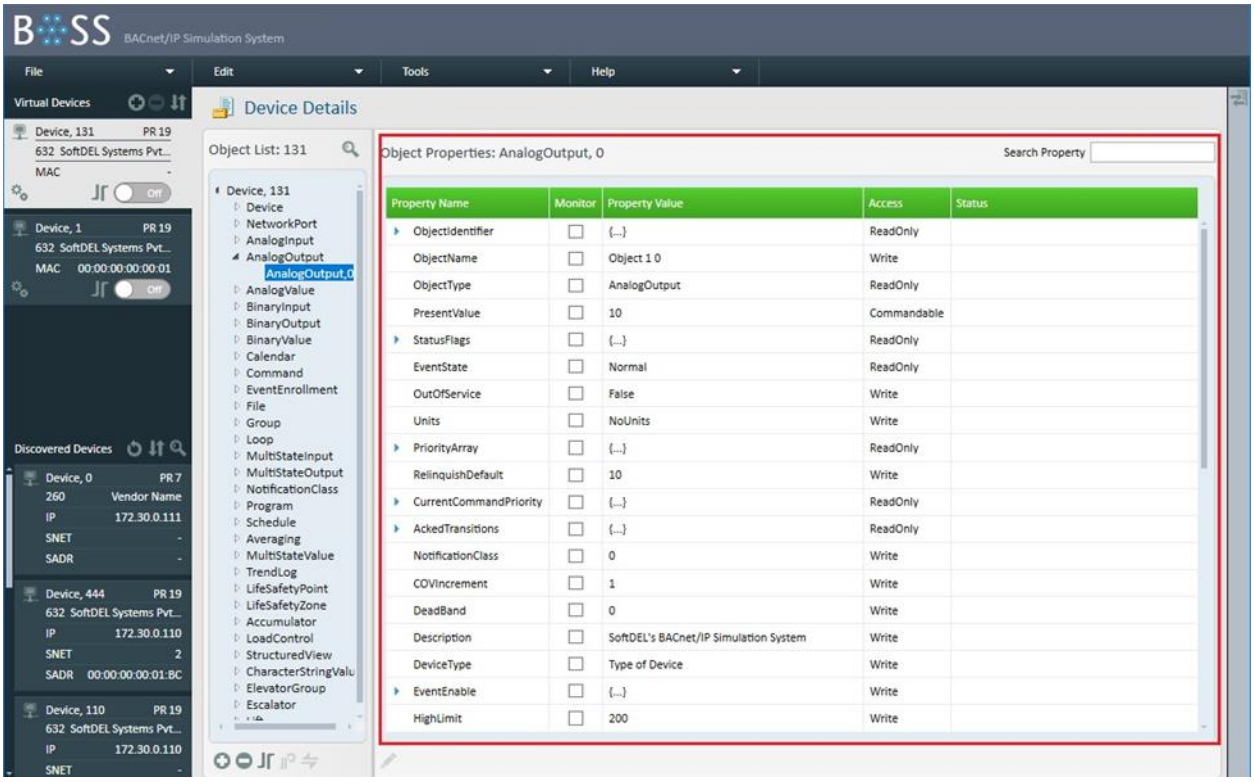


Object Properties Window Pane

When user selects any object from object list window, the **Object Properties** window displays list of properties and property values of selected object. Properties for which 'unknown-property' error code is received are not displayed, if the object belongs to discovered device.

While reading properties using RPM-A service, fallback mechanism may be used in following scenarios:

1. Reject is received with reason 'unrecognized-service' or 'buffer-overflow'
2. Abort is received with reason 'segmentation-not-supported' or 'buffer-overflow'



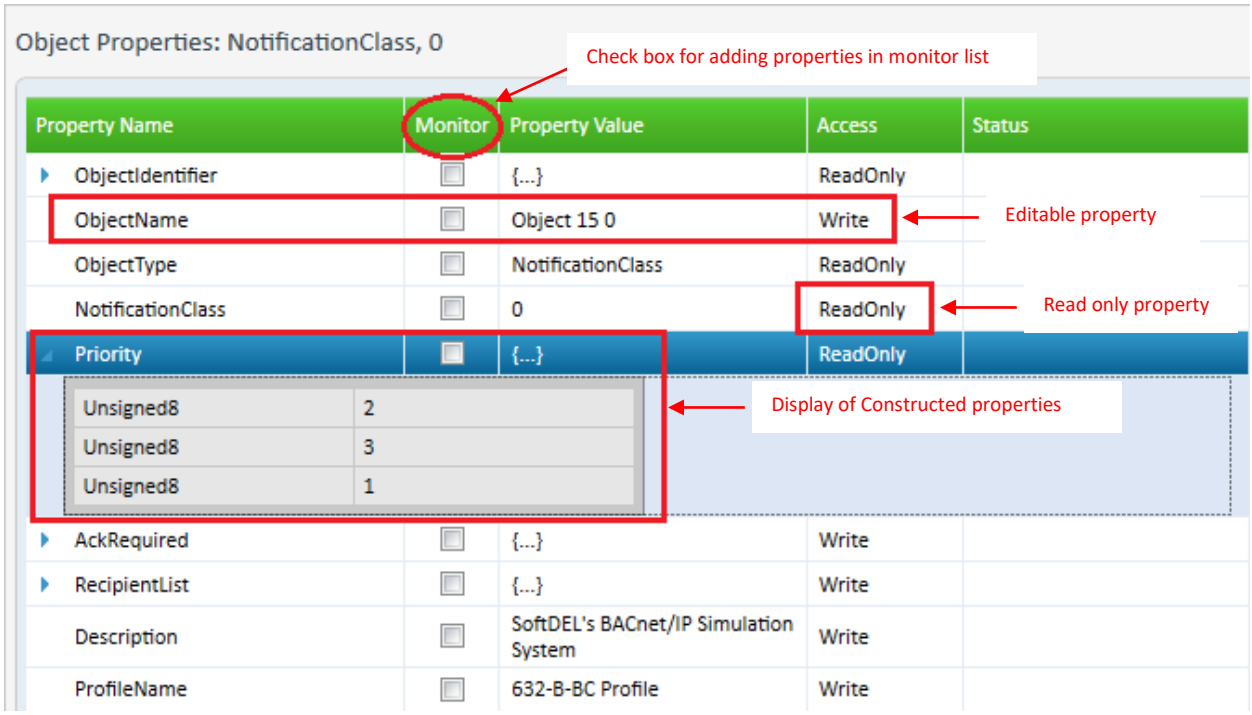
Object properties window displays the information in following format.

1. Property Name

Property name displays the name of property. This column expands and collapses node for constructed properties.

2. Property Value

Property Value displays value of the respective property.



3. Access Type

This column displays the access level of the property. These can be one of the following:

Access Type	Description
Read Only	Implies the property value is read-only by default and access type cannot be changed if property is required or mandatory
Read	Implies the property value is set to read by the user and access type can be changed
Write	Implies the property value is writable and access type can be changed
Commandable	Implies the property is commanded via priority-array property and access type cannot be changed
Not Supported	Implies the property access is set to not-supported or unknown by user and access type can be changed

Table 5: Object Properties Pane Access Type

Notes:

Access type can be changed by clicking on respective properties. This column is not displayed for object properties of discovered devices.

Changing the access type of one property will not affect access type of any other dependent property. E.g. if access type of 'Change-Of-State-Count' is set as 'not supported' or 'read', access type of 'Change-Of-State-Time' will not be changed.

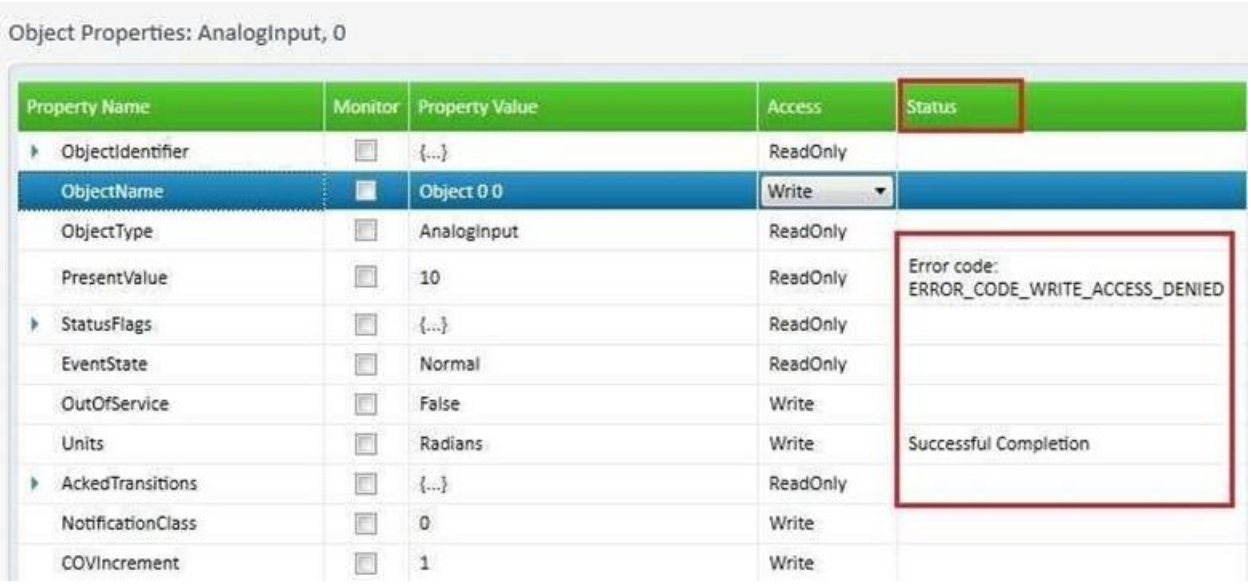
Access type of properties may change at runtime based on property dependencies. E.g. access type of 'Reliability' property changes as per value of 'Out-Of-Service' property.

Access type of properties mentioned below are set to their default values, in replicated objects or when project file is opened.

- Object Name (all object types)
- Change of State Count (binary objects)
- Elapsed Active Time (binary objects)
- Record Count (Trending objects)
- File Size (File object)

4. Status

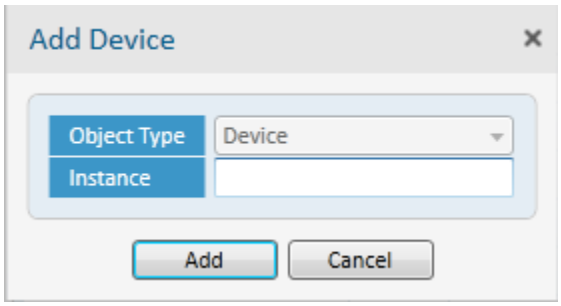
This column displays status of read or write property request. User can view response and different error codes received while reading or writing the property.



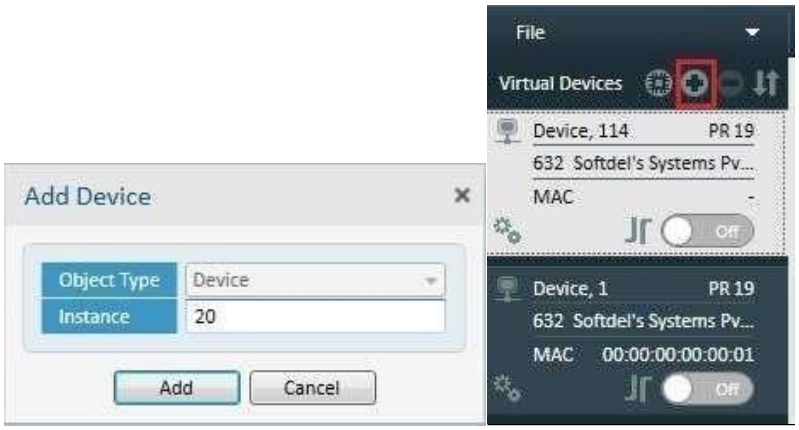
6 Add, Delete and Replicate Virtual Devices

6.1 Addition of Virtual Device

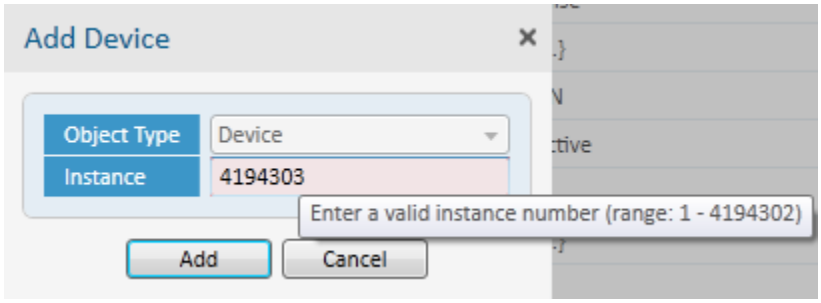
Click on Add Device icon present on Virtual Device windowpane, which opens the following window:



Once user enters valid instance number and clicks on **Add** button, device gets added into virtual device list as shown below.



Valid range for device instance number is from 1 to 4194302. When user enters invalid instance number; message tool tip gets displayed describing valid range.



6.2 Removal of Virtual Device

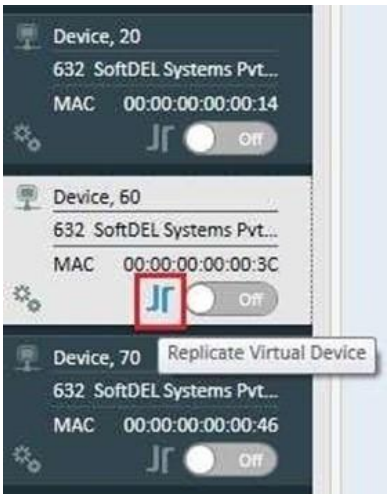
Select the virtual device to be removed and click on icon present on Virtual Device window.

Note:

User can't remove the default device (1st device created when the application started) present in the virtual device list.

6.3 Replicate Virtual Device

With this option user can create single or multiple copies of existing virtual device. Replication adds all existing objects and instances of selected virtual device to new virtual device.

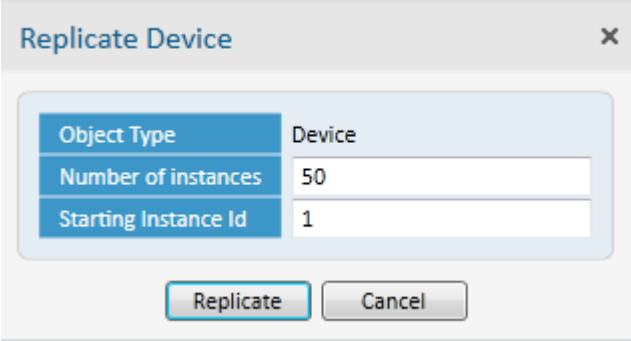


Select the device, that user wants to replicate and then click on icon.

E.g. To replicate (device, 60), click on icon present on (device, 60)

Following pop up window appears on screen. Enter valid number of instances, starting instance ID and click on **Replicate** button.

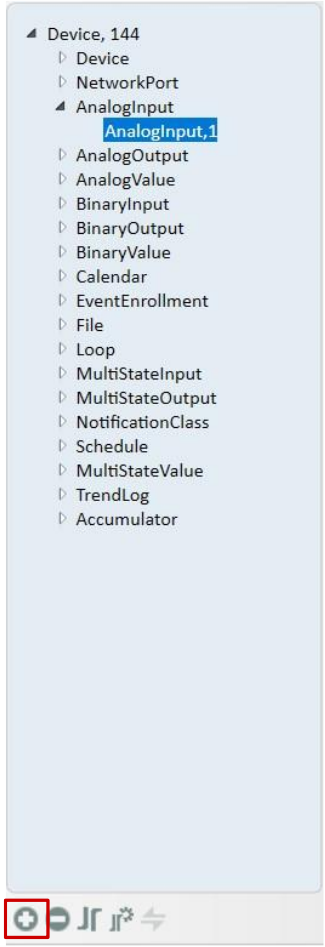
New device(s) with given instance numbers is/are added to virtual device list.


Replicate Device
Object Type
Device
Number of instances
50
Starting Instance Id
1
Replicate
Cancel

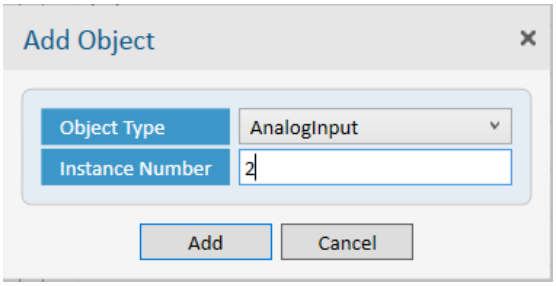
7 Add, Delete and Replicate Objects

7.1 Addition of Object

To add new object, click on icon present at the bottom of object list windowpane.



Following pop up window appears on screen:

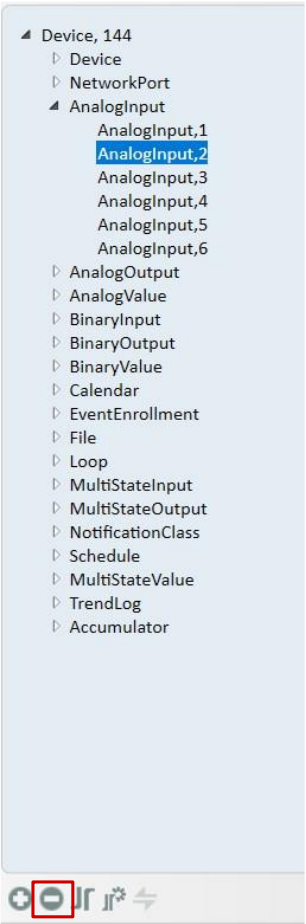


Select object type from the drop down list, enter valid instance number then click on **Add** button.

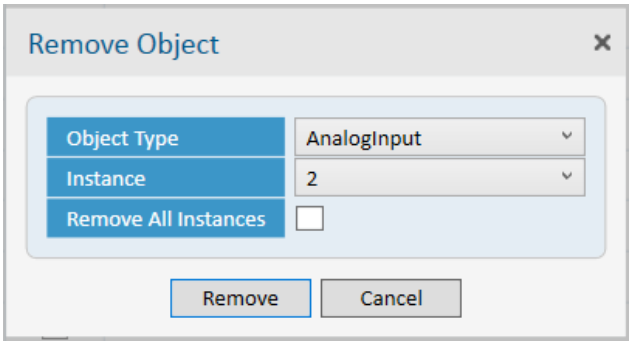
The object list is refreshed and new object appears in object list.

7.2 Removal of Object

To remove any object present in virtual devices, click on icon present at the bottom of object list windowpane.



Following pop up window appears. Select object type and instance number of the object to be removed and click on icon.

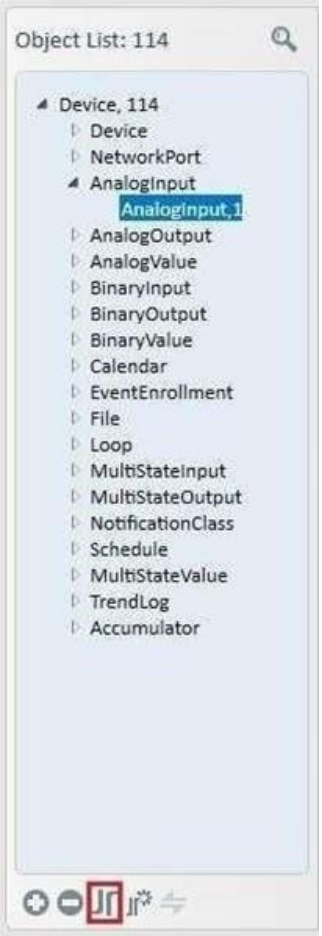


If all instances of specific object type need to be removed then, check **Remove All Instances** option.

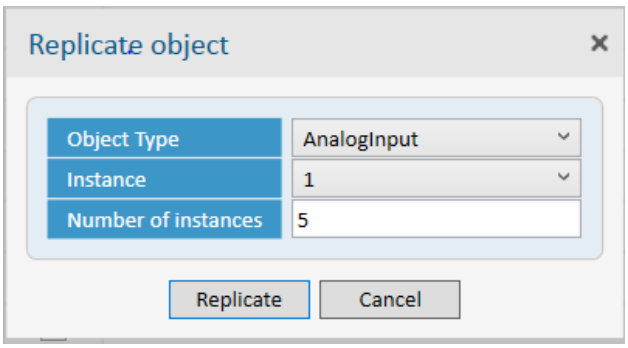
The object list is refreshed and deleted object(s) disappears from object list.

7.3 Replication of Object Instance

To replicate object instance, select the object that is to be replicated and click on icon.



Following window appears on screen:



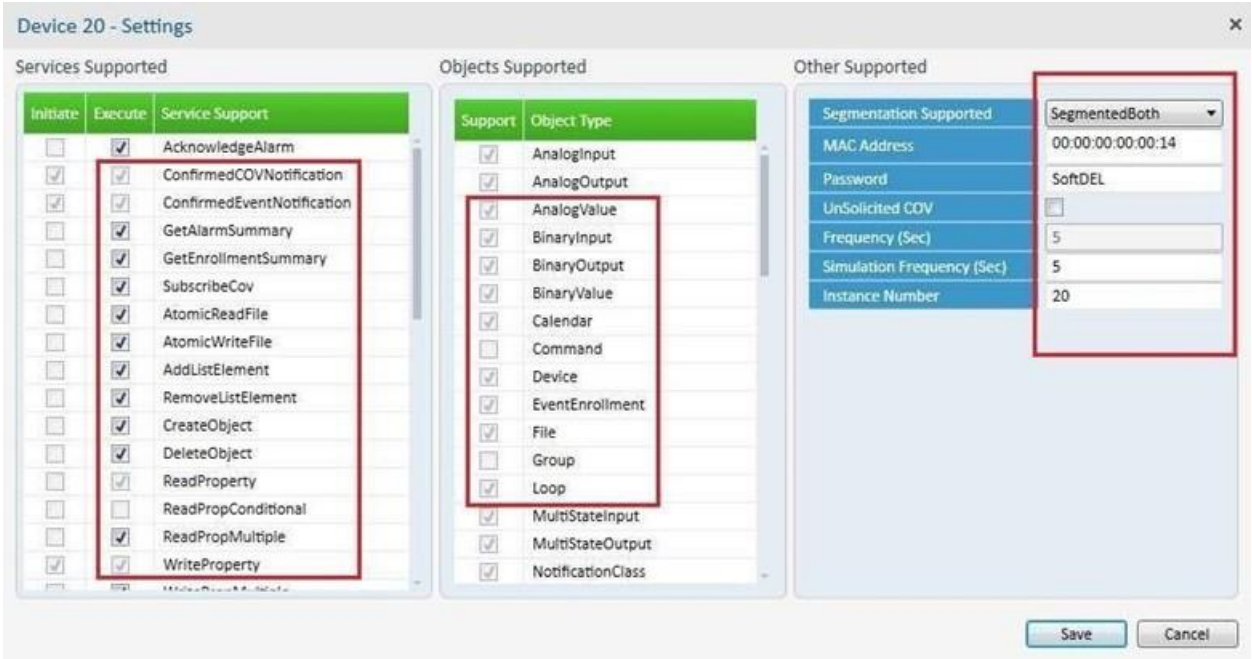
Select the object type and enter the number of instances to replicate, and then click on **Replicate** button.

The object list is refreshed, and new object(s) appear in object list.

8 Device Configuration

Users can view and modify configuration of virtual device with the help of **Device Settings** button.

Following window appears.



The device settings window allows the user do the following configurations:

1. Services Supported

It displays the A-side and B-side services supported by device.

A-side service support cannot be configured, whereas B-side service support can be configured.

2. Objects Supported

It displays the different object types supported in the simulator. These settings cannot be changed.

3. Segmentation Support

Users can change the segmentation support of respective virtual device from this window.

4. Password

The device password can be changed from this window. The maximum character length of the password field is 20. The password is used for DCC and RD services.

5. MAC address

Each virtual device is assigned a distinct MAC address. The MAC address is 6 bytes long and is not user configurable.

6. Simulation frequency

Simulation frequency can be changed from this window.

Minimum Simulation Frequency	1 second
Maximum Simulation Frequency	86400 seconds

Default value of Simulation frequency is 5 seconds.

More information is provided in 'Simulation' chapter.

7. Unsolicited COV Frequency

The Unsolicited-COV functionality support can be enabled by checking the checkbox next to the Unsolicited-COV frequency field as shown below.

Unsolicited-COV frequency can be changed from this window.

Minimum Unsolicited Frequency	1 second
Maximum Unsolicited Frequency	86400 seconds

Default value of Unsolicited-COV frequency is 5 seconds.

Note that by default unsolicited COV of each object instance within respective virtual device is OFF, even if 'UnSolicited COV' checkbox has (✓) mark. User can turn ON unsolicited for each object instance using **Unsolicited COV settings** icon at the bottom of object list window pane.

8. Instance Number

Users can change instance number of device using this option. Note that this configuration is applicable to only 1st virtual device in BOSS.

After setting all above configurations, click **Save** button to save the settings.

9 Simulation

Simulation feature is provided to simulate the present values of following object types:

1. Analog Objects
2. Binary Objects
3. Multi-State Objects.
4. Accumulator Object

Above objects are referred to as primary objects' in this section.

Table below provides information on simulation types supported by primary objects.

#	Object Type	Simulation Types Supported
1	Analog (Input, Output, Value)	Random, Ramp
2	Binary (Input, Output, Value)	Random, Reverse
3	Multi-State (Input, Output, Value)	Random, Ramp
4	Accumulator	Random, Ramp

Table 6: Simulation Types Supported By Respective Object Types

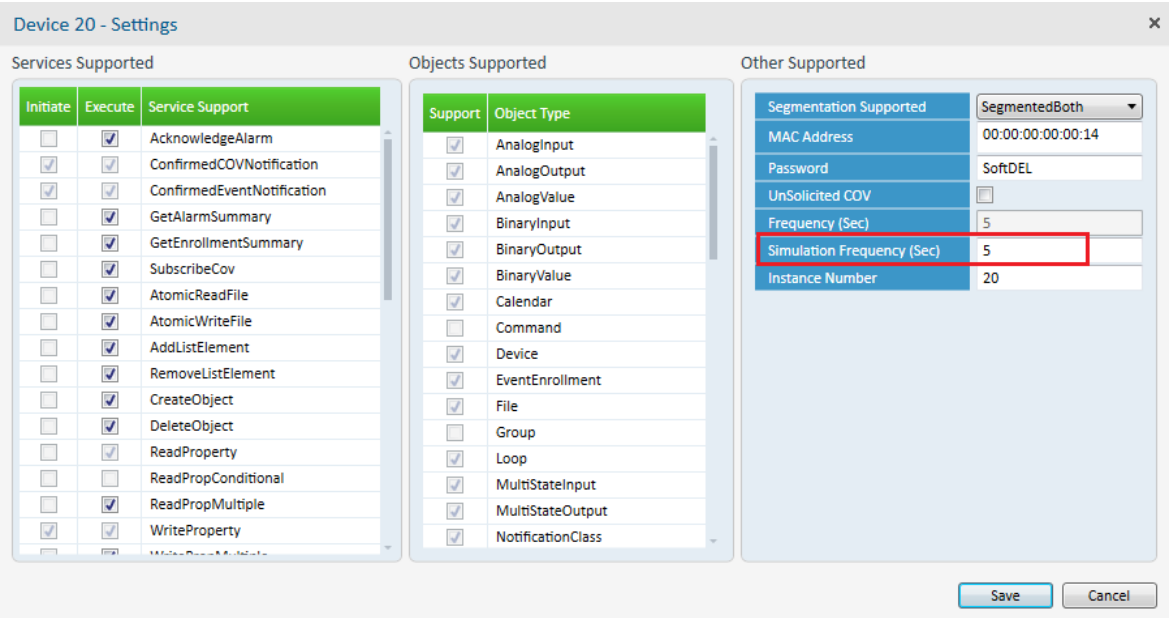
Default simulation type is 'Random'. For 'Ramp' simulation type, increment can be specified. The range for ramp simulation type is 1 to 65535.

9.1 Start Simulation

Simulation can be started by following process,

- Device Level: Simulation for all the supported object type instances will start at once at a device level.
- Object Type Level: Simulation for selected object type instances will start at object level.

Configure the simulation feature by clicking on **Device setting** icon which is present in left side of virtual device list. This will open the Device Setting window.



9.2 Simulation Frequency

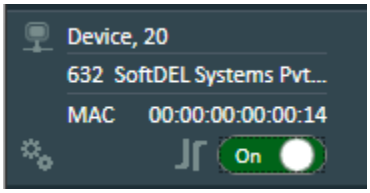
Simulation frequency can be changed from device setting window.

Minimum Simulation Frequency	1 second
Maximum Simulation Frequency	86400 seconds

Default value of Simulation frequency is 5 seconds.

Click on

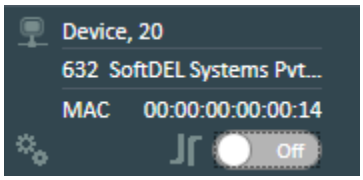
ON/OFF button to start simulation. Once the simulation feature is turned ON, the button changes to GREEN color.



By default simulation of all primary objects is turned ON, with default settings. Once the user starts device simulation, present values of primary objects get simulated.

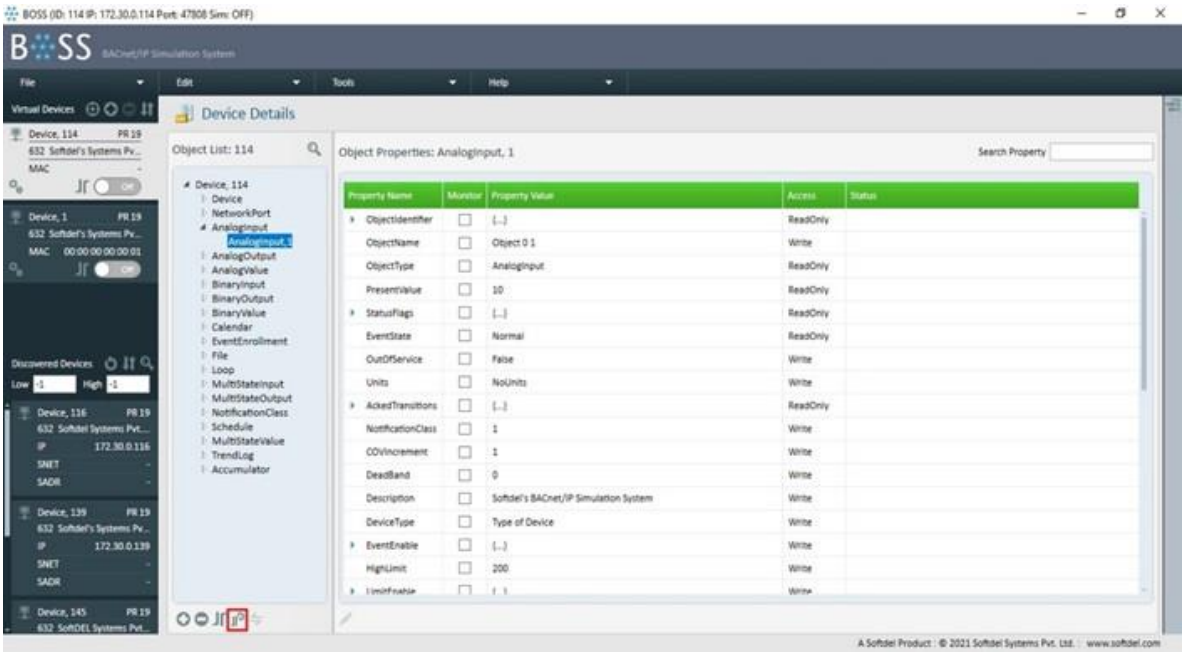
User can stop simulation feature of the device by clicking the

button, once started. After the simulation feature is turned OFF, the button changes to GRAY color.

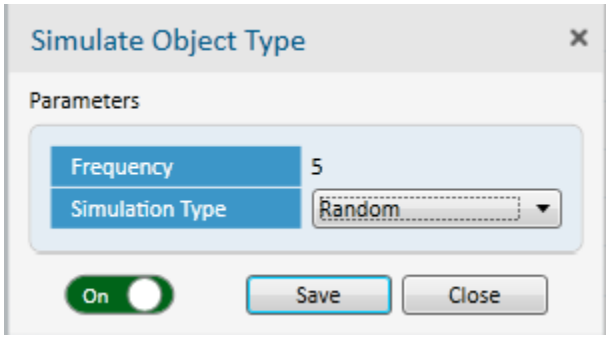


Once the user stops device simulation, present values of primary objects stop simulating.

Select the object instance or object type from object list of virtual device and then click on **Simulate Object Type** icon. The icon is present at the bottom of the main window under object list windowpane.



Clicking on **Simulate Object Type** icon will open following pop up window. User can change the simulation settings for respective object type. These settings are applied to all object instances present within selected object type. Using this window user can change **Simulation Type** and **Simulation ON/OFF** status for selected object type.



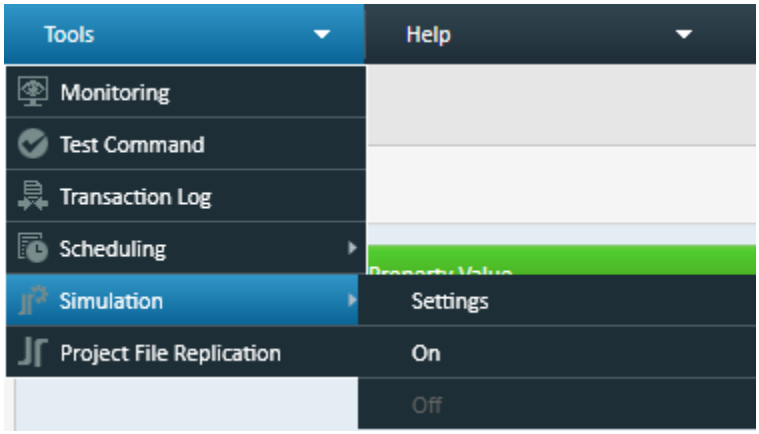
Note that during simulation:

- Present-Value is written or updated even if Out-Of-Service property value is false.
- Present-Value is written or updated at highest priority within Priority-Array.

Above two operations are applicable to respective object types, as per BACnet standard.

9.3 Global Simulation Settings

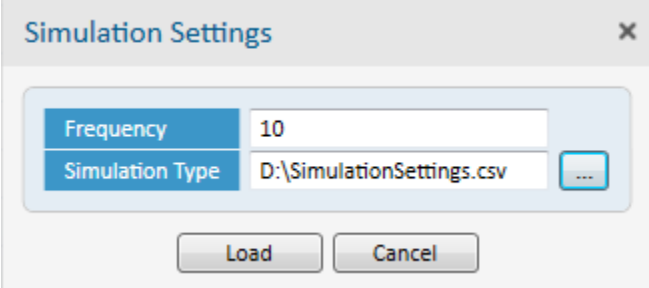
Global simulation settings can be used when user wants to start or stop simulation of all devices with single click. To use this setting go to **Tools->Simulation** menu.



User can also change default frequency and default simulation type (as per object type) using **Tools->Simulation->Settings** options. When simulation is turned ON, these settings will be applied to all devices.

Using following **Simulation Settings** window, user can set **Frequency** and **Simulation Type** (as per object type).

Frequency is a mandatory field whereas simulation type is optional. If simulation type is not specified, default settings are applied.



Once loaded the settings are saved till respective BOSS instance is running.

To start simulation of all devices, select **Tools->Simulation->On** option.

To stop simulation of all devices, select **Tools->Simulation->Off** option.

'Off' option is enabled only if simulation of at-least one device is turned ON.

'On' and 'Settings' options are enabled only if simulation of all devices is turned OFF.

Note that global simulation settings are not applied when simulation is turned ON using

i.e. ON/OFF button of respective device.

9.3.1 Input CSV File Format

Global simulation settings use input CSV file for loading object type wise simulation type and increment for ramp simulation type.

Parameter names for **ObjectType** and **SimulationType** should be same as provided in the example below. However, data entered in CSV file is case in-sensitive and spaces between parameters will be ignored.

Note that line(s) starting with 2 stars i.e. ** in these files will be considered as a comment.

Example:

**ObjectType, SimulationType, IncrementValue

AnalogInput, Ramp, 5

AnalogOutput, Random

AnalogValue, Random

BinaryInput, Reverse

BinaryOutput, Random

BinaryValue, Reverse

MultiStateInput, Random

MultiStateOutput, Ramp, 10

MultiStateValue, Random

Accumulator, Ramp, 15

Note: For Analog objects, simulation functionality will not work if these objects don't support “MaxPresValue” and “Min-Pres-Value” properties.

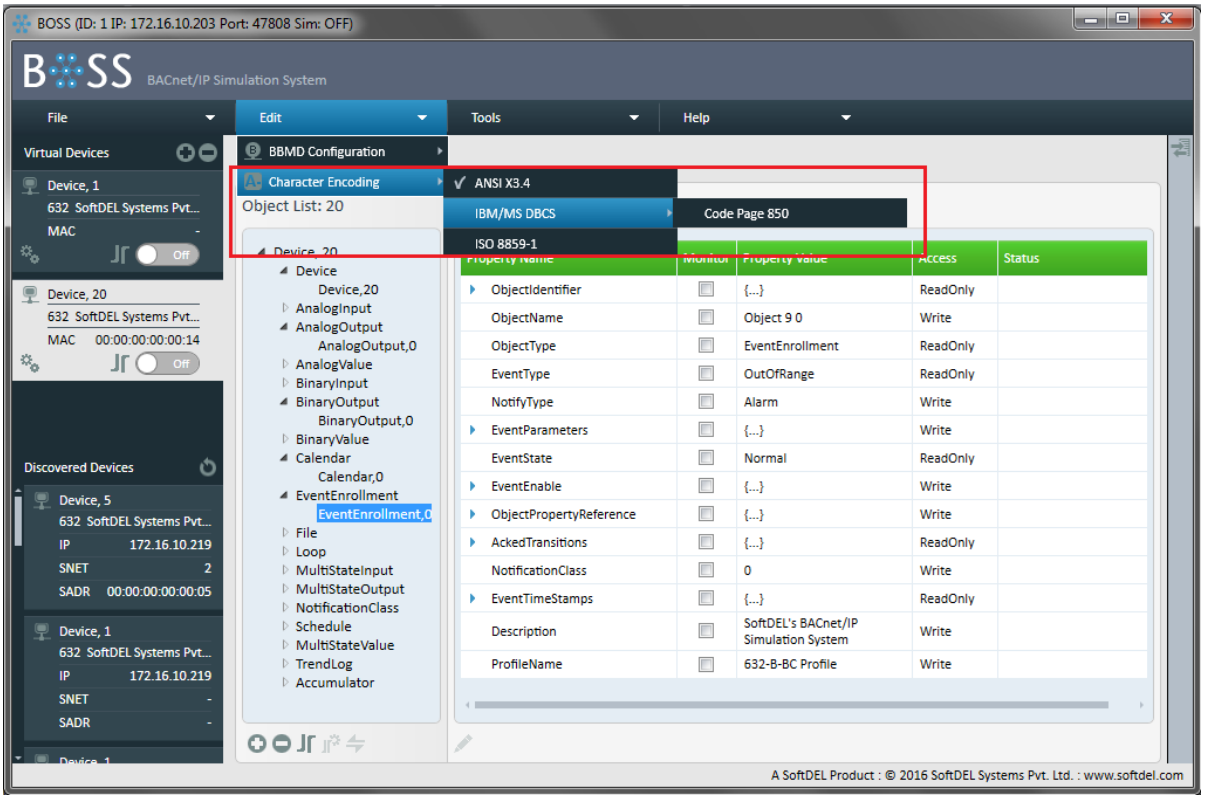
10 Character Set Support

BOSS supports the following three different types of character encoding.

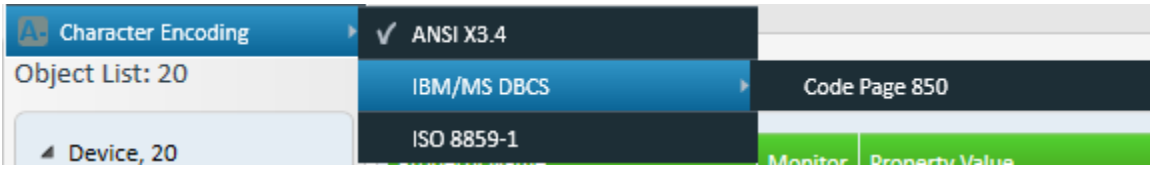
1. ANSI X3.4
2. IBM-MSDBCS (code page 850)
3. ISO 8859-1

By default, ANSI X3.4 character encoding is supported.

In order to change the character encoding, go to **Edit->Character Encoding** menu**.**



Select one of the options that is displayed, as shown below.

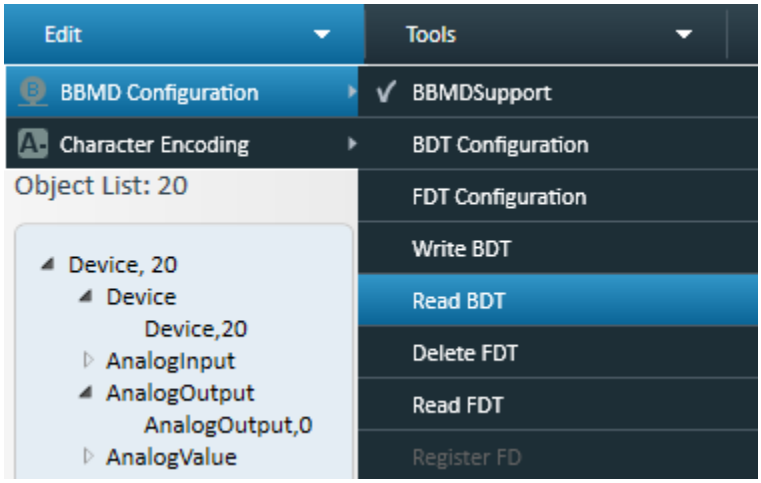


11 BBMD/ Non-BBMD Functionality

11.1 BBMD mode

To enable/disable the BBMD support of BOSS, check '**BBMD Support**' option from **Edit** Menu on the main window.

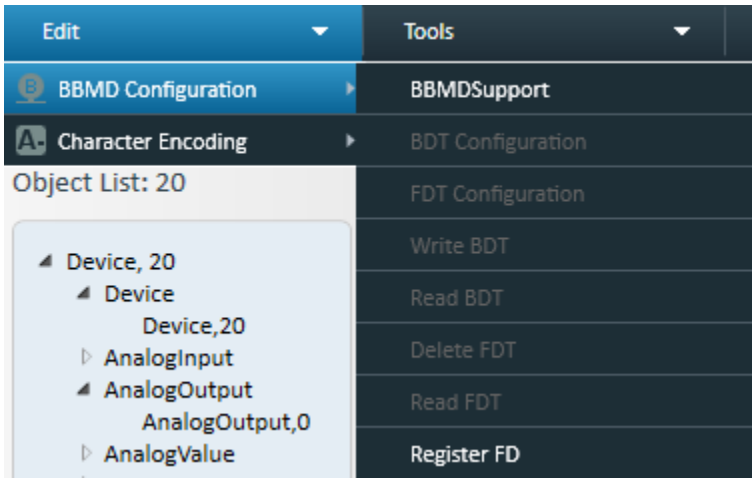
When user checks 'BBMD Support' BOSS acts like BBMD device on network and other BBMD configurations are enabled.



When user reopens BOSS, then BBMD Support displays status configured during last session. Note that while performing '**BDT Configuration**', one BDT entry of BOSS itself should be added.

11.2 Non-BBMD mode

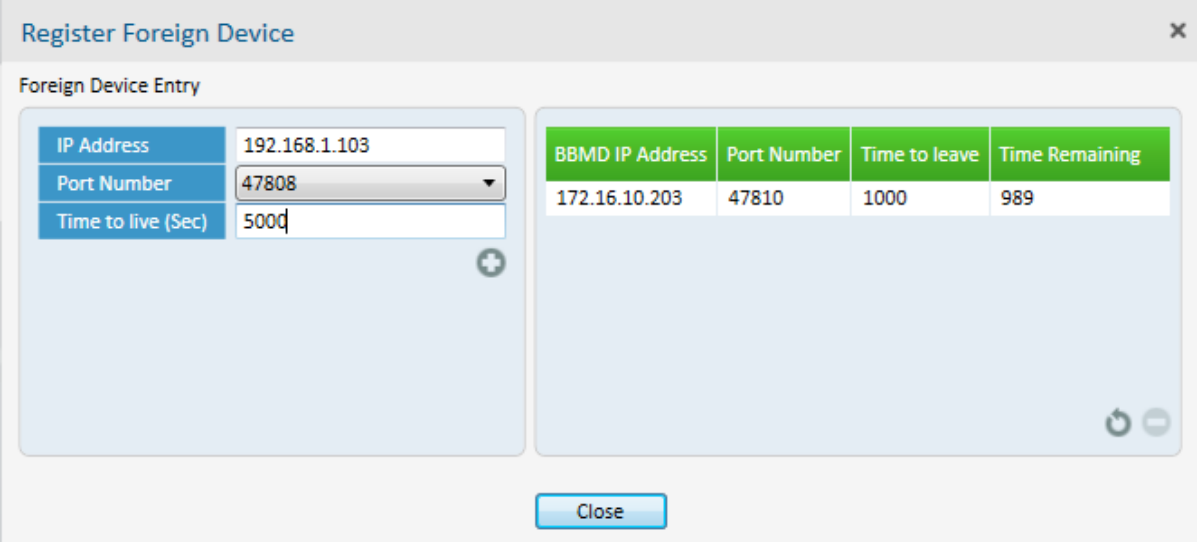
When 'BBMD Support' is unchecked BOSS acts like Non-BBMD device on network.



11.3 Register Foreign Device

Register Foreign Device option is used to initiate register foreign device request to any accessible BBMD device, present in the internetwork. Note that BBMD support should be disabled to use this option.

When user clicks on **Register FD** sub menu, following window appears.



IP Address: The IP Address of BBMD device to whom the registration has to be done.

Port No: The Port No of BBMD device for which the registration has to be done.

Time to live (seconds): Specify how long the device should be registered as Foreign Device.

Click on 'Save' button. Register BOSS as FD with other BBMD devices on internetwork.
Once registration is successful, FD entries will be displayed in right portion of the window.

12 Auto Device Discovery

Purpose of "Auto Device Discovery" is to automatically discover all the BACnet devices available on network.

12.1 Device Discovery

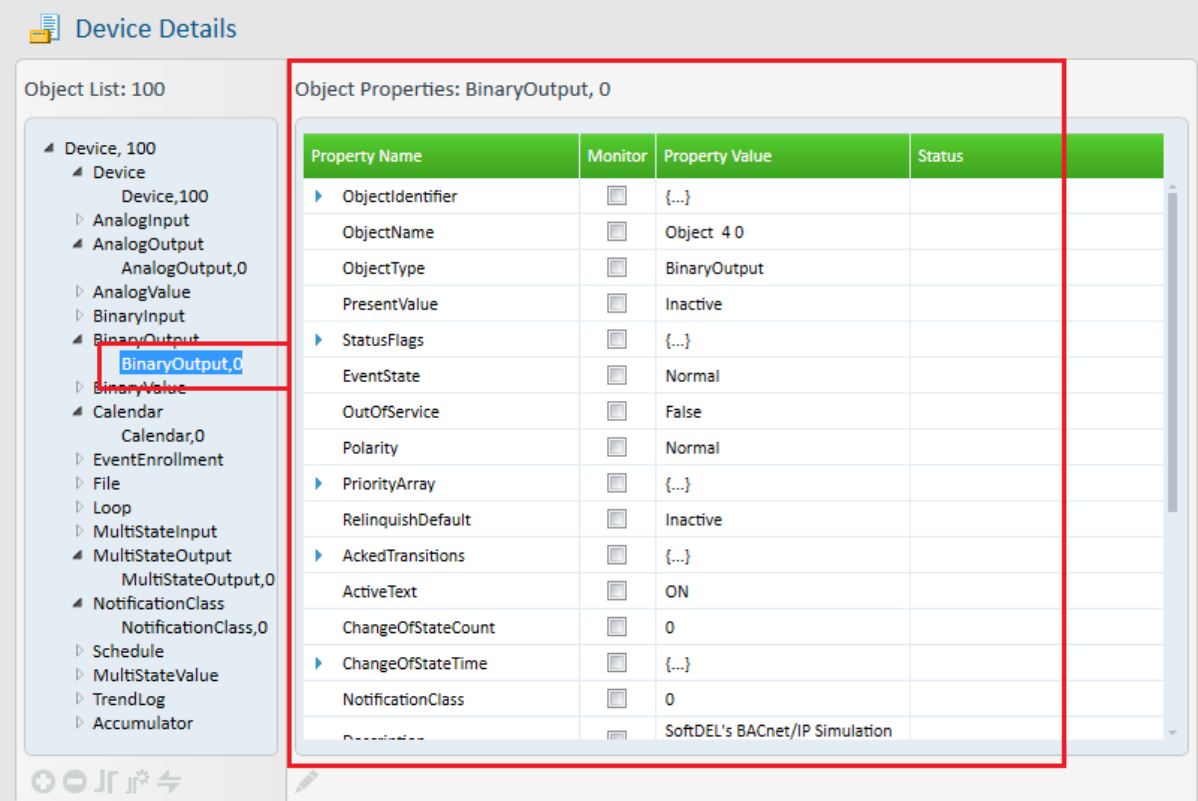
List of discovered devices is displayed in the bottom left of main window. User can refresh the list by clicking **Refresh** button.

12.2 Object Discovery

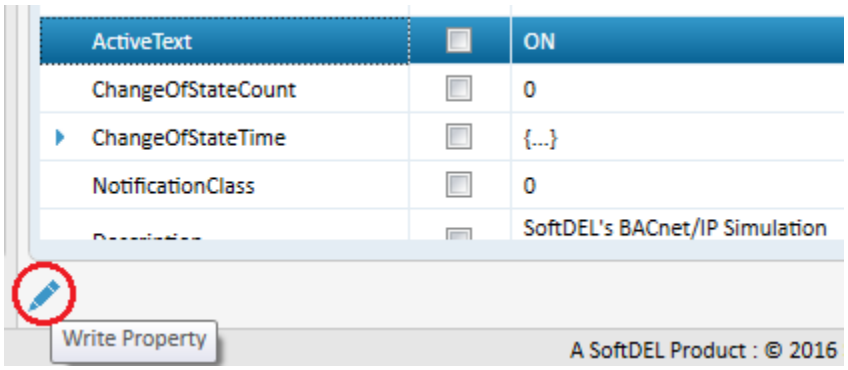
Select a device to read its object list.

12.3 Property Discovery

Select object from the object list to view or read properties present within the object.



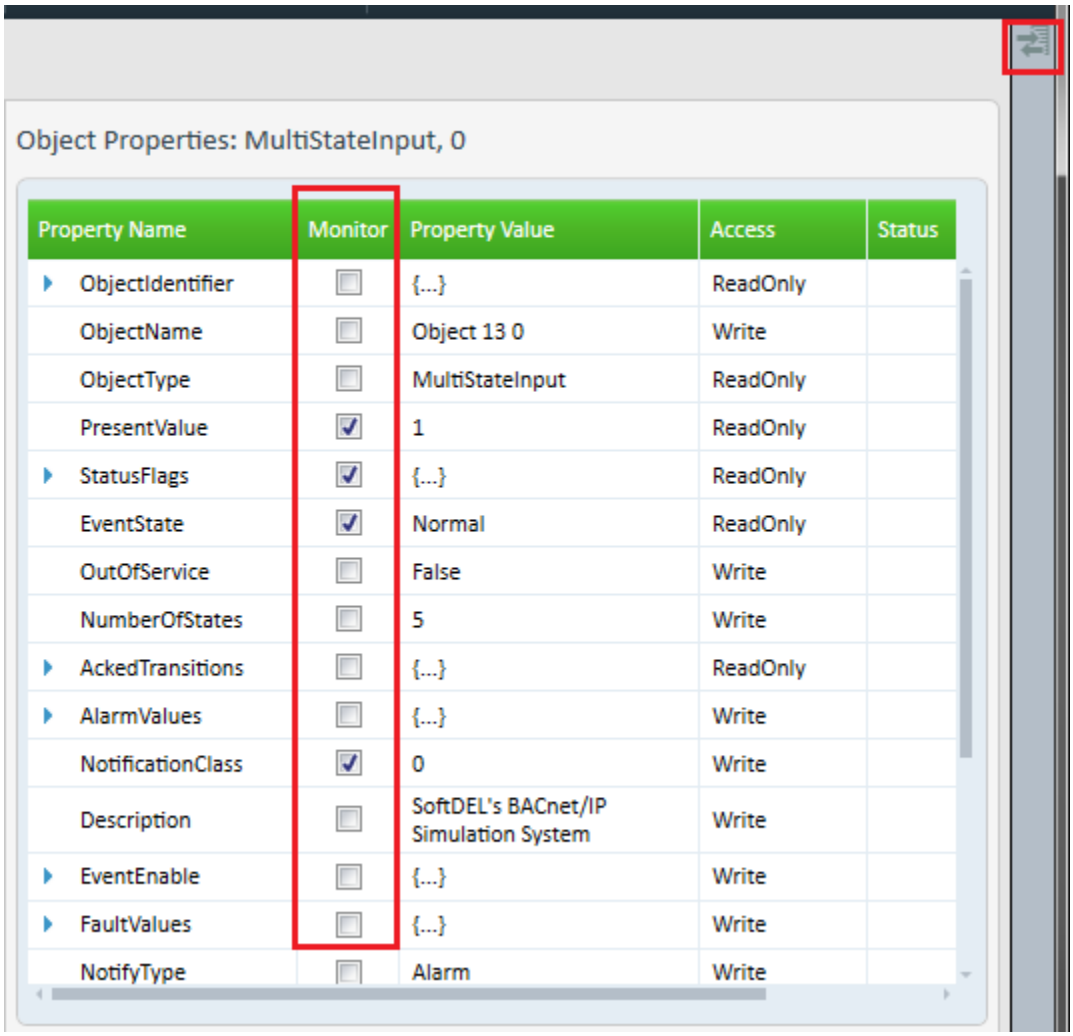
User can write property of any object by selecting the property and clicking on **Write Property** button present at the bottom. This will open the write property window. Double click on any property, will also open the write property window.



13 Monitor Properties

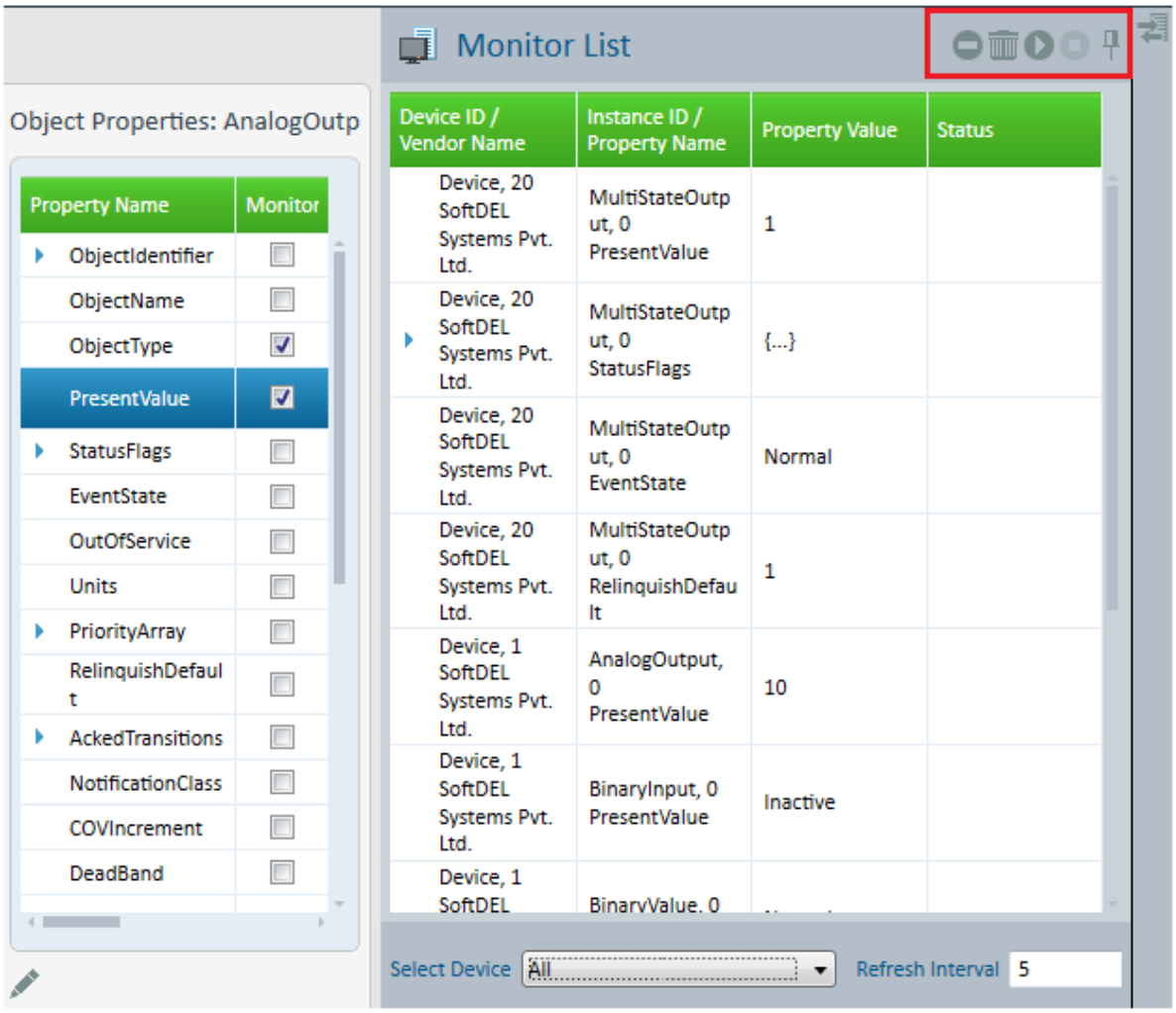
User can monitor different properties from the same object, multiple objects or even from multiple devices at same time.

Click (✓) on the check box, to monitor property value of any specific property. To open monitor list window click on **Monitor list** icon present at the right hand side as shown below. User can also open the same from **Tools->Monitoring** menu.



Once user clicks on the monitor list icon the Monitor list window (following window) appears. The window has option to dock or undock.

Monitor list window shows list of all the properties selected for monitoring.



Following actions can be performed using various icons present at top right corner of Monitor List window

1. To start monitoring the property values, enter the refresh interval and click on **Start** button.

2. User can stop monitoring by clicking **Stop** button.

3. To remove any property from monitor list, select the property and click on **Remove Selected** button or uncheck the Monitor check box in the "Object Properties" windowpane.

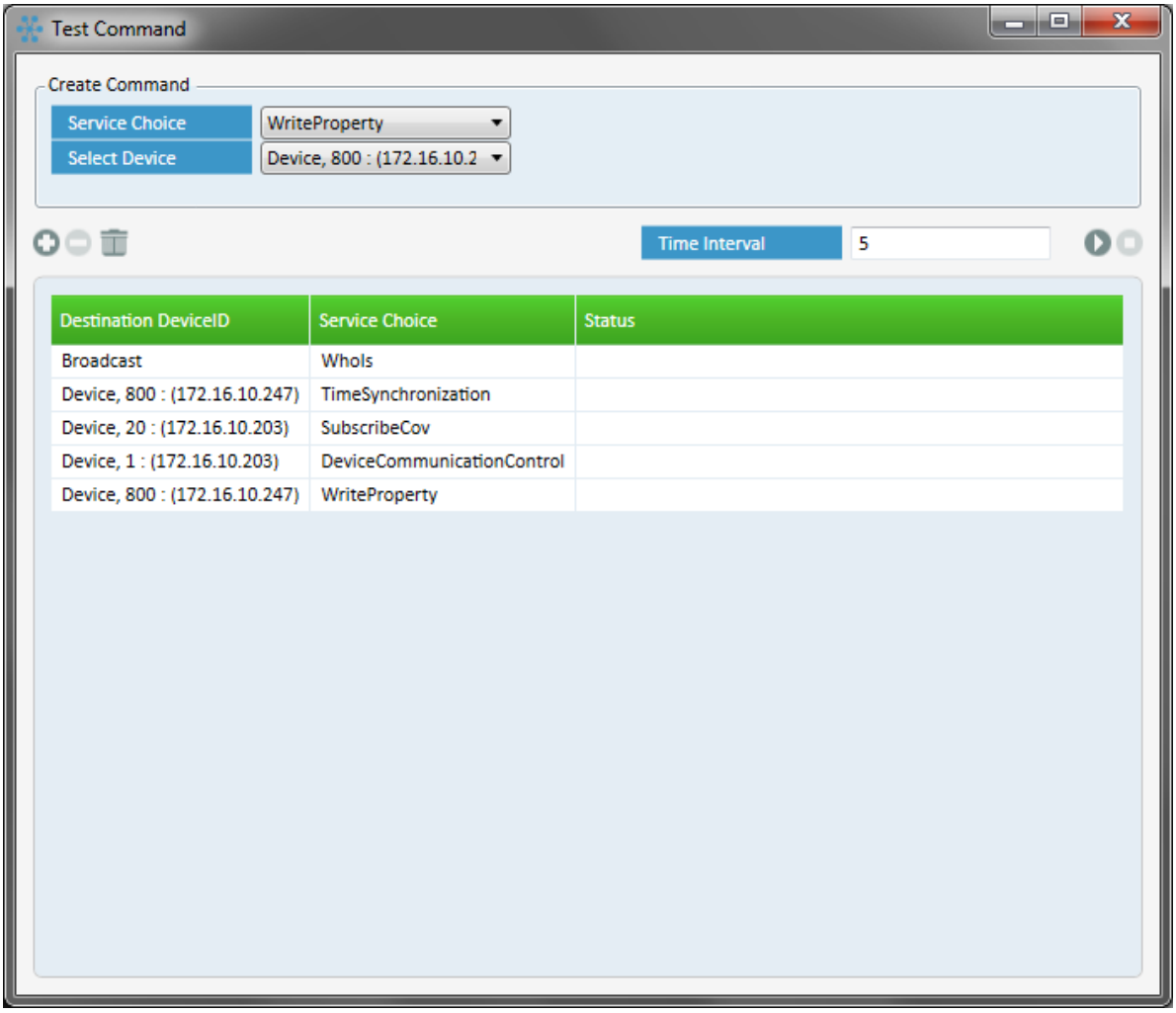
4. **Clear list** button clears all the properties from monitor list window.

5. To dock-undock Monitor List window to Main window use button.

14 Test Command

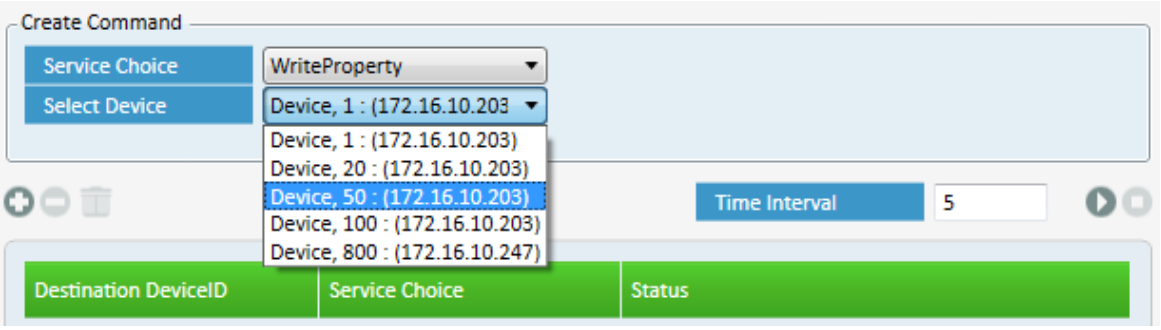
Test command is used to send A-side service request(s) to devices present in the network. The commands are sent continuously as per specified interval.

To open test command window go to **Tools->Test Command** menu.



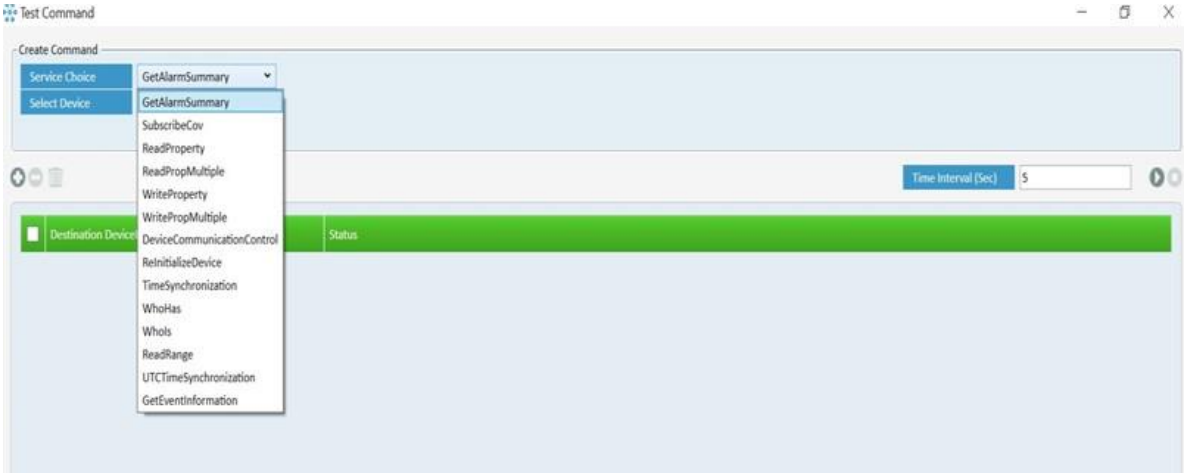
Step 1:

To add test command, select the device from the select device drop down list.



Step 2:

Select the service to be sent from the service choice drop down list.



Step 3:

After selecting device and service choice, click on **Create Command** button.

Step 4:

Fill the request data for selected service and click **Add** button. Command will be added to the list. User can add multiple commands to the list.

After adding the commands, enter time interval to periodically send the service request to destination device.

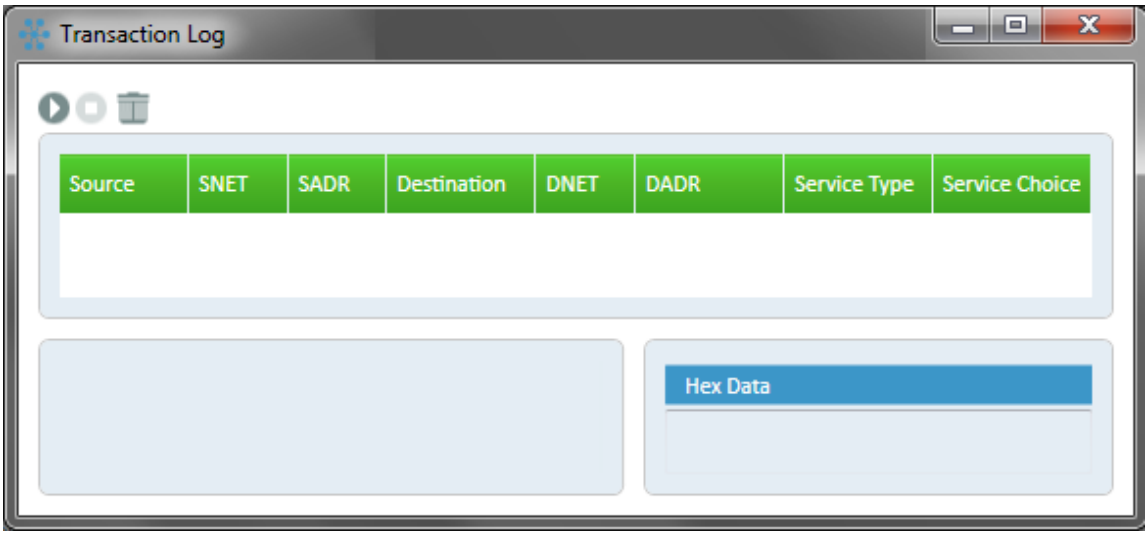
Click on **Start** button to start the process of sending commands.

Click **Stop** button to stop sending test commands

Select specific test command from the list and click **Remove Selected** button to remove selected test command from the list. Click **Clear list** button to empty test commands list.

15 Transaction Log

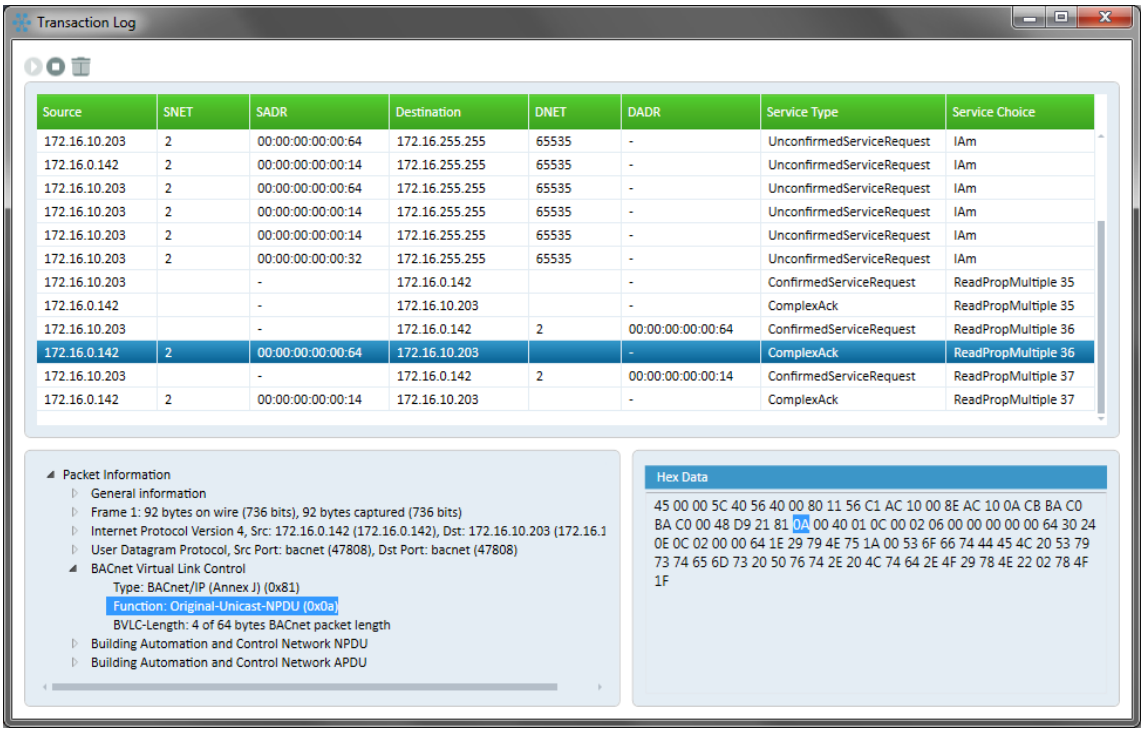
To open transaction log window go to **Tools->Transaction Log** menu. Following screen appears.



Click on **Start** button to start capturing data, sent to or received from network.

Click **Stop** button to stop capturing data, sent to or received from network.

Click **Clear list** button to clear all captured data, sent to or received from network.



On selecting any row of captured data, packet information and HEX data is displayed in above window. The packet information and HEX data is displayed in bottom left windowpane and bottom right windowpane respectively.

Note that to use transaction log functionality, T-SHARK or Wireshark should be installed in the system. Also, administrative privileges are required to run BOSS.

16 Customer Support

For any information related to BOSS or related support, kindly write to boss-support@softdel.com or info@softdel.com

17 References

ANNEX A – Running BOSS using CLI

BOSS can be started using "Windows Command Prompt" through BOSS applications CLI i.e. command line interface.

Command Line Parameters

Command line parameters supported by BOSS are as shown in below table:

#	Parameter	Keyword	Description
1	IP Address	/IP: or /ip:	IP address to be used for BACnet communication
2	UDP Port Number	/PORT: or /port:	UDP port number to be used for BACnet communication
3	Device ID	/DEVID: or /devid:	Device ID to be assigned to first device i.e. pure IP device created in respective BOSS instance.
4	Project File	/PROJFILE: or /projfile	BOSS project file to create devices (in BOSS) with different configurations
5	Simulation Operation	/SIM: or /sim:	Simulation ON-OFF operation to be applied. Default value is OFF.
6	Simulation Frequency	/SIMFREQ: or /simfreq:	Simulation frequency to be applied. Default value is 5 seconds,
7	Simulation Type	/SIMTYPE: or /simtype:	Simulation type to be applied. Default values are same as provided in simulation chapter.
8	Device Discovery	/DEVDISC: or /devdisc	Device Discovery ON-OFF operation to be applied. Default value is ON
9	Schedule File	/SCDFILE: or /scdfile:	Master schedule file to be used for scheduling from CSV.
10.	Simulation Priority	/SIMPRIORITY: or /simpriority	Default simulation priority to be set for Present Value write operation during simulation and scheduling from CSV.

Table 7: BOSS CLI Parameters

/IP, **/DEVID**and **/PORT** command line parameters are required. **/PROJFILE, /SIM, /SIMFREQ, /SIMTYPE**, **/DEVDISC, /SCDFILE** and **/SIMPRIORITY** command line parameters are optional. Parameter keywords are case in-sensitive. Double quotes ("") are required if the path mentioned in **/PROJFILE** command line parameter contains a 'space' character. There should be no 'space' character between command line keyword and parameter value.

Using CLI

To start BOSS from command prompt, use following steps:

1. Open "cmd" i.e. command prompt with administrative privileges.
2. Go to "C:\Program Files\SoftDEL\BOSS\" directory. This is the default BOSS installation directory. The path may be different, if changed during BOSS installation.
3. Type "**BOSS.exe /IP:172.16.10.164 /DEVID:23 /PORT:47808 /PROJFILE:D:\BOSS\Projects\projectfilename.bosp /SIM:ON /SIMFREQ:10 /SIMTYPE:D:\BOSS\Projects\SimulationSettings.csv /DEVDISC:OFF /SCDFILED:\ BOSS\Projects\Master1.csv /SIMPRIORITY:10"** to run single instance of BOSS.

4. To create multiple BOSS instances, execute step 3 above (multiple times) with unique device-id and unique combination of IP-address and UDP-port.

Note:

User should provide correct parameter values in the format same as shown above in step number 3. Simulation settings are applied after loading the specified BOSS project file.

ANNEX B – Running Multiple Instances of BOSS on Single Computer

Use BOSS shortcut created in **Windows -> Start** menu or desktop to launch multiple instances of BOSS on a single computer.

BOSS's command line interface can also be used to launch multiple instances of BOSS.

Note:

1. When launching multiple BOSS instances, if the input parameters are invalid, respective error will be displayed on BOSS UI.
2. Number of BOSS instances that can be created on a single computer will depend on system resources e.g. RAM, processor speed.

Pre-Requisite

On a single computer, multiple instance of BOSS should be created with:

1. Device Id which is unique on respective computer. Ideally the device ID should be unique throughout the BACnet internetwork.
2. Combination of IP-Address and UDP port which is unique.

Automating Multiple BOSS Instances Creation

Creation of multiple BOSS instances can be automated using a small Windows batch (.bat o .BAT) file. The script to be written in batch file is as given below.

```
cd \ cd "C:\Program Files\SoftDEL\BOSS\" start BOSS.exe
/IP:172.16.10.164 /DEVID:23 /PORT:47808  start BOSS.exe
/IP:192.168.1.105 /DEVID:27
/PORT:47809
/PROJFILE:"D:\BOSS\Projects\projectfilename.bosp" pause
```

Above batch file script will start two instances of BOSS. Two devices (one in each BOSS instance) will be created on respective BOSS instance.

Note that the batch file should be run with administrative privileges.

Log File

For error analysis, separate log file will be created for each instance. Name of the log file will be a combination of device-id and the time-stamp at which respective BOSS application was started.

E.g. LOG_DEVICE_ID-15_2015-10-29_19-51-48.txt

Known Limitations

List of functionalities that will not work properly (or may behave abnormally) if multiple instances of BOSS are created on a single computer, is mentioned in following table:

#	Functionality	Description
1	Transaction Log	Transaction log functionality described in this document
2	BBMD	BBMD functionality described in this document
3	File Object	File object property linking or dependency

#	Functionality	Description
4	Atomic Read File – B	Atomic read file B-side service
5	Atomic Write File – B	Atomic write file B-side service
6	Backup Restore – B	Backup and restore B-side functionality

Table 8: BOSS Multiple Instances Known Limitations

Linking i.e. B-side dependency between 'Align-Interval' and 'Interval-Offset' properties of device object is not supported.

ANNEX C – Custom Scheduling Using Input CSV Files

Custom scheduling functionality can be used to simulate values of certain properties at specified time. The schedule is read from input CSV files.

Supported Objects and Properties

Table below provides information on object types and their properties supported in custom scheduling.

#	Object Type (Value)	Properties : Property Name (ID)
1	Analog Input (0)	Present Value (85), Out Of Service (81), Reliability (103)
2	Analog Output (1)	Present Value (85), Out Of Service (81), Reliability (103)
3	Analog Value (2)	Present Value (85), Out Of Service (81), Reliability (103)
4	Binary Input (3)	Present Value (85), Out Of Service (81), Reliability (103), Change Of State Count (15), Elapsed Active Time (33)
5	Binary Output (4)	Present Value (85), Out Of Service (81), Reliability (103), Change Of State Count (15), Elapsed Active Time (33), Feedback Value (40)
6	Binary Value (5)	Present Value (85), Out Of Service (81), Reliability (103), Change Of State Count (15), Elapsed Active Time (33)
7	Multistate Input (13)	Present Value (85), Out Of Service (81), Reliability (103)
8	Multistate Output (14)	Present Value (85), Out Of Service (81), Reliability (103), Feedback Value (40)
9	Multistate Value (19)	Present Value (85), Out Of Service (81), Reliability (103)
10	Accumulator (23)	Present Value (85), Out Of Service (81), Reliability (103), Value Set (191)

Table 9: Custom Scheduling Supported Objects And Properties

Note that during custom scheduling:

- Present-Value is written or updated even if Out-Of-Service property value is false.
- Present-Value is written or updated at highest priority within Priority-Array.

Above two operations are applicable to respective object types, as per BACnet standard.

Input CSV File Formats

Custom scheduling uses input CSV files of two different formats explained in this section. Note that line(s) starting with 2 stars i.e. ** in these files will be considered as a comment.

Master File

Only one master file is required for this functionality. Following information is obtained from master file:

1. Devices (IP device or virtual device) to be scheduled
2. Objects (type and instance) within the respective device to be scheduled
3. Pattern file(s) to be used, to schedule properties of respective object instances

The file format is as mentioned below:

** MASTER FILE

** DeviceID,ObjectType,ObjectID,FileName

Example:

10,1,10,“D:\BOSS\InputFiles\FileAnalog1.csv”

10,3,10,"D:\BOSS\InputFiles\FileBinary1.csv"

Details of each parameter within the master file are mentioned in table below:

#	Parameter Name	Supported Range	Description
1	DeviceID	0 to 4194302	Specifies device ID (IP or virtual) to which the object belongs.
2	ObjectType	0 to 127	Specifies object type.
3	ObjectID	0 to 4194302	Specifies object instance to be scheduled.
4	FileName	-	Specifies pattern file to be used for scheduling respective object. The file name should be provided in double quotes.

Table 10: Custom Scheduling Master File Parameters

All the parameters in this file shall be comma ',' separated.

Pattern File

One or many pattern file(s) are required for this functionality. Following information is obtained from pattern file:

1. Time at which scheduling is to be performed
2. List of property Id and property value at specified time

The file format is as mentioned below:

** PATTERN FILE

** Time,ListOfPropertValues(comma separated)

** HH:MM,(PropertyID=PropertyValue),(PropertyID=PropertyValue)

Example:

00:00,(85=12.23),(103=2)

23:59,(85=23.27)

Details of each parameter within the pattern file are mentioned in table below:

#	Parameter Name	Supported Range	Description
1	HH	0 to 23	Specifies time in hours, and uses 24 hour clock.
2	MM	0 to 59	Specifies time in minutes.
3	PropertyID	0 to 511	Specifies property ID of respective object.
4	PropertyValue	Any valid value are per property data type	Specifies property value of specified property.

Table 11: Custom Scheduling Pattern File Parameters

Time and list of property values shall be comma ',' separated. HH and MM shall be colon ':' separated. Property ID and property value shall be separated using equal to '=' sign. Property ID and property value pair shall be enclosed within opening bracket '(' and closing bracket ')'.

In pattern file, all the properties to be scheduled at a particular time (i.e. HH:MM) should be provided in single row.

Validations Supported

Validations supported in custom scheduling functionality are mentioned below.

1. Check for input file extension. Only CSV file format (.csv file extension) is supported.
2. Check for header in master file. Master file header is **** MASTER FILE**.
3. Check for header in pattern file. Pattern file header is **** PATTERN FILE.**
4. Check for 'DeviceID' and 'ObjectID' parameters in master file as per supported range i.e. 0 to 4194302.
5. Check for 'ObjectType' parameter in master file as per supported range i.e. 0 to 127.
6. Check for 'FileName' parameter in master file. File name or file name with file path must be in double ("") quotes.
7. Check for 'FileName' parameter in master file. If input file, with specified file name is not present at specified path, then error message is displayed.
8. Check for 'Time' i.e. 'HH:MM' parameter in pattern file as per supported range i.e. 0 to 23 and 0 to 59 respectively.
9. Check for 'PropertyID' parameter in pattern file as per supported range i.e. 0 to 511.
10. Check for strings and special characters in 'PropertyValue' parameter within pattern file.

Note that BOSS will skip the line, if no property-id and property-value pair is provided after time. Also, validations other than those mentioned above, are not performed on input data. When incorrect input data is provided by user, the tool may behave in unspecified manner. Following are examples of unspecified behavior:

1. Operation involving incorrect data is not performed
2. All other operations following incorrect data are not performed

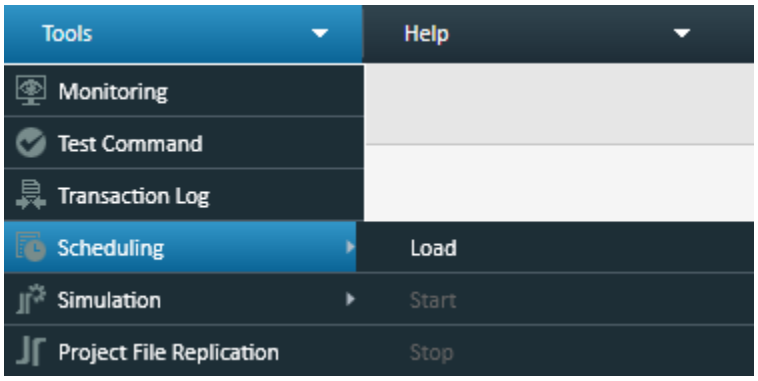
In such cases, user can identify errors by analyzing logs (if available), actual behavior and input data. Incorrect data should not cause BOSS to crash.

Load, Start and Stop Scheduling

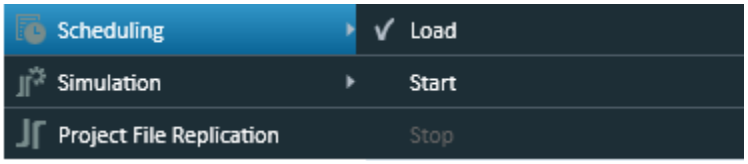
Custom scheduling functionality is divided in 3 different operations, these are as follows:

- Load – to load scheduling data from input files
- Start – to start scheduling
- Stop – to stop scheduling

The scheduling options can be accessed from tools menu i.e. from **Tools->Scheduling** menu.



On successfully loading all input CSV files, 'CSV files loaded successfully' dialog box appears and (✓) will be displayed in front of Load option.



When load operation is performed, previously loaded data if present will be deleted and then new data will be saved.

If load operation is unsuccessful, error message dialog box appears indicating line number of file in which error occurred during load operation.

Start option is enabled only if previous load operation was successful. Stop option is enabled only if scheduling is started. Load option is disabled if scheduling is currently in progress i.e. started. Start and stop options are disabled if load operation is unsuccessful or not performed.

Miscellaneous

Values supported by 'Reliability' property are provided in table below:

#	Reliability Value	Reliability Specifies
1	0	No fault detected
2	1	No sensor
3	2	Over range
4	3	Under range
5	4	Open loop
6	5	Shorted loop
7	6	No output
8	7	Unreliable other
9	8	Process error
10	9	Multi state fault
11	10	Configuration error
12	11	Reserved by standard
13	12	Communication failure
14	13	Member fault

Table 12: Custom Scheduling Reliability Property Values

Note that not all values will be supported by each object type. For details, refer description of 'Reliability' property for respective object type, provided in BACnet standard.

Values supported by 'Reliability' property as per object type are provided in table below:

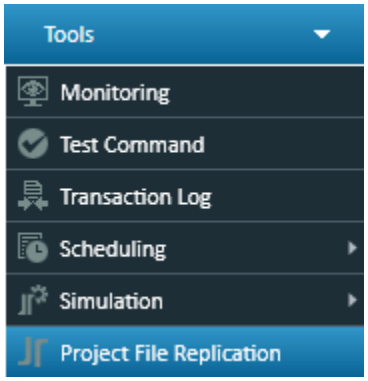
#	Object Type	Reliability Property Value
1	Analog Input	No fault detected, No sensor, Over range, Under range, Open loop, Shorted loop, Communication failure, Unreliable other
2	Analog Output	No fault detected, Open loop, Shorted loop, No output, Communication failure, Unreliable other

#	Object Type	Reliability Property Value
3	Analog Value	No fault detected, Over range, Under range, Communication failure, Unreliable other
4	Binary Input	No fault detected, No sensor, Open loop, Shorted loop, Communication failure, Unreliable other
5	Binary Output	No fault detected, No output, Open loop, Shorted loop, Communication failure, Unreliable other
6	Binary Value	No fault detected, Communication failure, Unreliable other
7	Multistate Input	No fault detected, No sensor, Over range, Under range, Open loop, Shorted loop, Multi state fault, Communication failure, Unreliable other
8	Multistate Output	No fault detected, Open loop, Shorted loop, No output, Communication failure, Unreliable other
9	Multistate Value	No fault detected, Multi state fault, Communication failure, Unreliable other
10	Accumulator	No fault detected, No sensor, Over range, Under range, Open loop, Shorted loop, Communication failure, Unreliable other

Table 13: Custom Scheduling Reliability Property Values As Per Object Type

ANNEX D – Project File Replication

Project file replication functionality can be used to create one or more copies of input BOSS project file. These copies are created with device ID value(s) specified by user. To access this functionality go to **Tools** menu and select **Project File Replication** option.

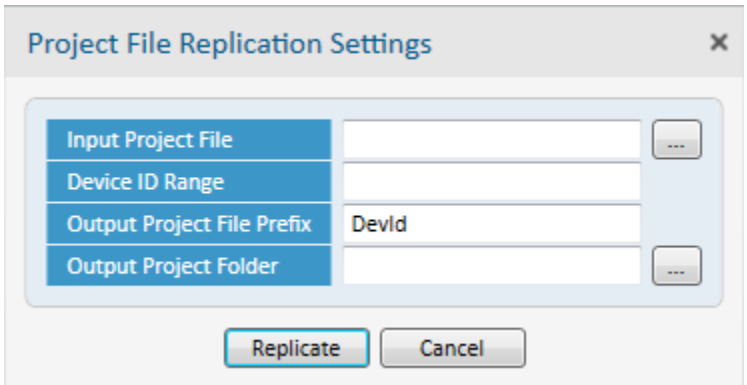


Pre-Requisite

The project file replication functionality works only on input project file that contains information of only one device i.e. while saving the project file from file menu, only 1 virtual device should be present.

Settings

Following popup appears when user clicks on 'Project File Replication' option.



Input Project File: Path of source or input project file can be provided using the browse button.

Device ID Range: Range of device IDs to create one or more copies, can be provided in this text box. User can use hyphen '-' and comma ',' to provide multiple ranges. Example: **1-10, 25, 30-100, 101, 102**

Output Project File Prefix: Output project file(s) are created with name having the prefix provided in this text box. This prefix can be of maximum 10 characters. Only alphabets, numbers, underscore '_' and hyphen '-' are allowed in prefix.

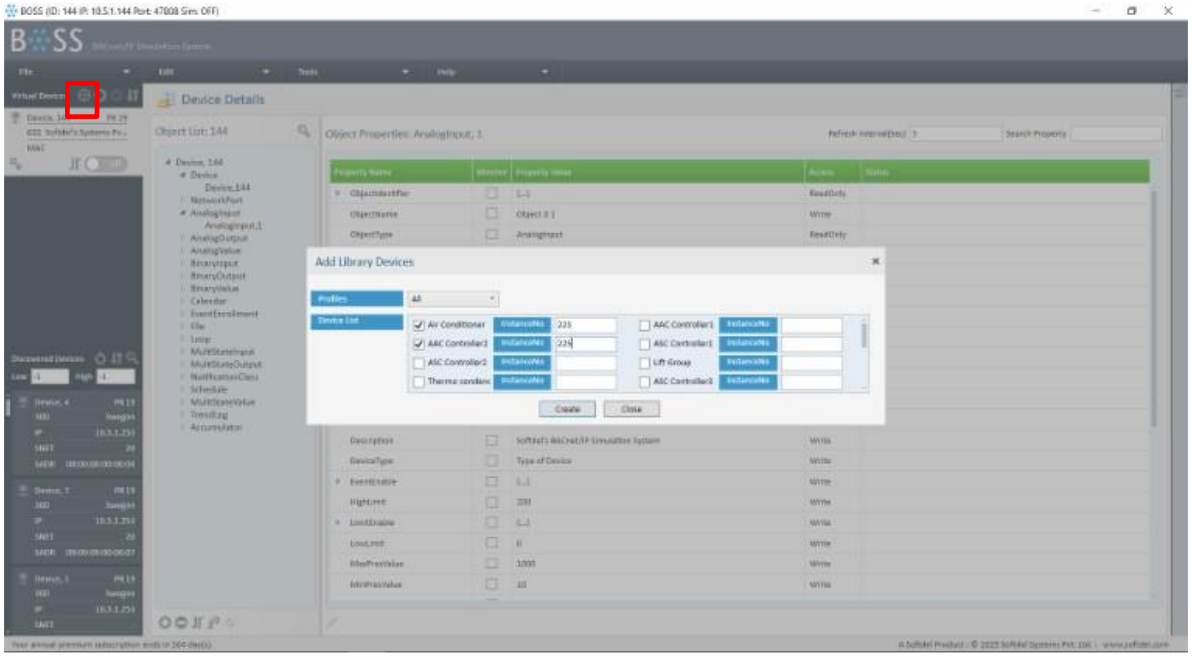
Output Project Folder: Path of folder to save output project file(s) can be provided using the browse button.

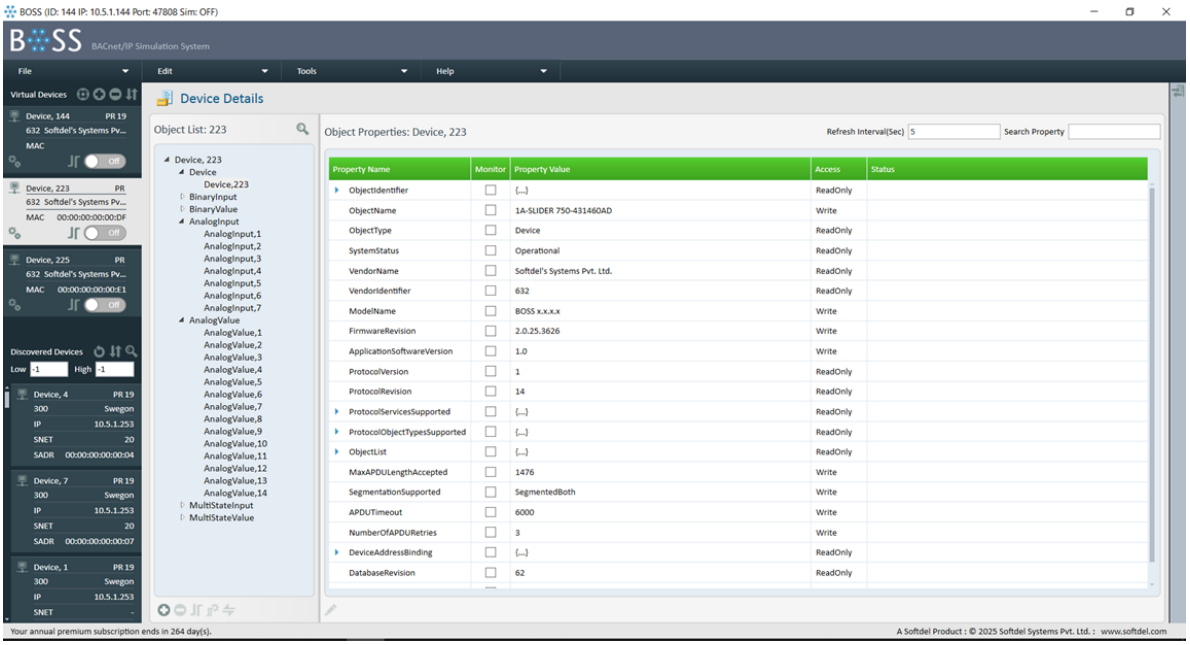
ANNEX E – Library Devices

Library Devices add functionality can be used to create virtual device(s) based on profiles. This feature provides a set of predefined devices.

To access this functionality, follow the below steps:

1. Click on below shown icon. Add Library Devices user interface (UI) popped up.
2. Select the profile. Based on selected profile, Device(s) available in Device List section.
3. Checked the required device and provide the device instance number.
4. Click on “Create” button.





ANNEX F – Read CSV File and Create Device

BOSS has ability to read *.bosp project file and create device(s) based on the previous data stored within the file. Read CSV functionality enables BOSS to read *.csv/*.CSV project file and create device(s). CSV files can be read using command line interface of BOSS or from **File->Open** menu.

Device objects are created with device instance numbers present in CSV file name. Device object properties are initialized with default value. Properties of other object types are initialized with value from input CSV file. If property value is not present in CSV file, respective property is initialized with default value.

The sample CSV files are present in folder "SampleCSVFile" at the installed path.

CSV File Format Details

The CSV file contains details of object type and its properties value for one device. Note that the CSV file doesn't support creation of multiple devices.

1. First row (i.e. Headline) is used to show the definition information of a column which is corresponding to the relevant BACnet property identifier.

   Note that first row starting with hash character i.e. # in CSV file will be considered as headline.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	#75	77	79	-1	117	65-1	69	72	45	59	46	4	84	43	74	110-1	110-2	110-3	110-4	11
2	1	AI1	0	1	5	1000	0	0	4	3										
3	4194305	AO1	1	1	5	1000	0	0	4	3										

2. The column and the associated property ID on first row are fixed.
3. Each row (except first row) with valid object type and its properties value is considered for object instance creation in device.

CSV File Name format

CSV file name should be "DEV" + Device Instance No. (7 digits representing base 10 value, i.e. 0 to 4194302) + "V" + Version No. (2 decimals) + ".csv".

Example: DEV0011001V02.csv

DEV0011001V02.csv

Supported Objects and Properties

Table below shows details of objects and its properties supported for reading values from input CSV file.

#	Object Name	Property Name	Property ID	Conformance Code
1	Analog Input (0)	Object-Identifier	75	R
2		Object-Name	77	R
3		Object-Type	79	R
4		Units	117	R
5		Max-Pres-Value	65	O
6		Min-Pres-Value	69	O
7		Notify-Type	72	O
8		High-Limit	45	O

#	Object Name	Property Name	Property ID	Conformance Code
9		Low-Limit	59	O
10		Event-Enable	35	O
11		COV-Increment	22	O
12		Event-Detection-Enable	353	O
13		Notification-Class	17	O
14	Analog Output (1)	Object-Identifier	75	R
15		Object-Name	77	R
16		Object-Type	79	R
17		Units	117	R
18		Max-Pres-Value	65	O
19		Min-Pres-Value	69	O
20		Notify-Type	72	O
21		High-Limit	45	O
22		Low-Limit	59	O
23		Event-Enable	35	O
24		COV-Increment	22	O
25		Event-Detection-Enable	353	O
26		Notification-Class	17	O
27	Analog Value (2)	Object-Identifier	75	R
28		Object-Name	77	R
29		Object-Type	79	R
30		Units	117	R
31		Notify-Type	72	O
32		High-Limit	45	O
33		Low-Limit	59	O
34		Event-Enable	35	O
35		COV-Increment	22	O
36		Event-Detection-Enable	353	O
37		Max-Pres-Value	65	O
38		Min-Pres-Value	69	O
39		Notification-Class	17	O
40	Binary Input (3)	Object-Identifier	75	R
41		Object-Name	77	R
42		Object-Type	79	R
43		Notify-Type	72	O
44		Inactive-Text	46	O

#	Object Name	Property Name	Property ID	Conformance Code
45		Active-Text	4	O
46		Polarity	84	O
47		Event-Enable	35	O
48		Event-Detection-Enable	353	O
49		Alarm-Value	6	O
50		Notification-Class	17	O
51	Binary Output (4)	Object-Identifier	75	R
52		Object-Name	77	R
53		Object-Type	79	R
54		Notify-Type	72	O
55		Inactive-Text	46	O
56		Active-Text	4	O
57		Polarity	84	O
58		Event-Enable	35	O
59		Event-Detection-Enable	353	O
60		Notification-Class	17	O
61	Binary Value (5)	Object-Identifier	75	R
62		Object-Name	77	R
63		Object-Type	79	R
64		Notify-Type	72	O
65		Inactive-Text	46	O
66		Active-Text	4	O
67		Event-Enable	35	O
68		Event-Detection-Enable	353	O
69		Alarm-Value	6	O
70		Notification-Class	17	O
71	Multi-State Input (13)	Object-Identifier	75	R
72		Object-Name	77	R
73		Object-Type	79	R
74		Notify-Type	72	O
75		Number-Of-States	74	R
76		State-Text	110	O
77		Event-Enable	35	O
78		Event-Detection-Enable	353	O
79		Alarm-Values	7	O
80		Fault-Values	39	O

#	Object Name	Property Name	Property ID	Conformance Code
81		Notification-Class	17	O
82	Multi-State Output (14)	Object-Identifier	75	R
83		Object-Name	77	R
84		Object-Type	79	R
85		Notify-Type	72	O
86		Number-Of-States	74	R
87		State-Text	110	O
88		Event-Enable	35	O
89		Event-Detection-Enable	353	O
90		Notification-Class	17	O
91	Multi-State Value (19)	Object-Identifier	75	R
92		Object-Name	77	R
93		Object-Type	79	R
94		Notify-Type	72	O
95		Number-Of-States	74	R
96		State-Text	110	O
97		Event-Enable	35	O
98		Event-Detection-Enable	353	O
99		Alarm-Values	7	O
100		Fault-Values	39	O
101		Notification-Class	17	O
102	Notification Class (15)	Object-Identifier	75	R
103		Object-Name	77	R
104		Object-Type	79	R
105		Notification-Class	17	O
106		Priority	86	R
107		Recipient-List	102	R
108	Schedule (17)	Object-Identifier	75	R
109		Object-Name	77	R
110		Object-Type	79	R
111		Priority-For-Writing	88	R
112	Calendar (6)	Object-Identifier	75	R
113		Object-Name	77	R
114		Object-Type	79	R
115	Accumulator (23)	Object-Identifier	75	R
116		Object-Name	77	R

#	Object Name	Property Name	Property ID	Conformance Code
117		Object-Type	79	R
118		Units	117	R
119		Notify-Type	72	O
120		Notification-Class	17	O
121		Max-Pres-Value	65	O
122		Scale	187	O
123		Event-Enable	35	O
124		Event-Detection-Enable	353	O
125	TrendLog (20)	Object-Identifier	75	R
126		Object-Name	77	R
127		Object-Type	79	R
128		Buffer-Size	126	R
129		Log-Interval	134	O
130	File (10)	Object-Identifier	75	R
131		Object-Name	77	R
132		Object-Type	79	R
133		File Type	43	R

Table 14: Read CSV Objects And Properties Details

Steps for CSV file preparation to create object with specified property value

This section explains the steps to configure BACnet objects and its properties value in CSV file.

Configure Property Values

Refer following tables for range value to configure object and its properties value in CSV file.

1. **Configure Object-Name, Object-Type, Object-Instance property:**

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	77	CharacterString	Analog Input (0) Analog Output (1) Analog Value (2) Binary Input (3) Binary Output (4) Binary Value (5) Calendar (6) Multi-State Input (13) Multi-State Output (14) Notification Class (15) Schedule (17) Multi-State Value (19) Accumulator (23)	Max 80 characters
2	79	Unsigned8		0 to 127
3	-1	-		0 to 4194302

Note: #75 is a dummy column value which is same as -1. But, it is not used while csv parsing.

2. **Configure Max-Pres-Value property:**

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	65-1	Real	Analog Input (0) Analog Output (1) Analog Value (2)	-2,147,483,648 to 2,147,483,647
2	65-2	Real	Accumulator (23)	-2,147,483,648 to 2,147,483,647

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	#75	77	79	-1	117	65-1	69	72	45	59	46	4	84	43	74	110-1	110-2	110-3	110-4
2	1	AI1	0	1	5	1000	0	0	4	3									
3	4194305	AO1	1	1	5	1000	0	0	4	3									
4	8388609	AV1	2	1	5	1000	0	0	4	3									
5	12582913	BI1	3	1				1			Disable	Enable	0						
6	16777217	BO1	4	1				0			Disable	Enable	0						
7	20971521	BV1	5	1				1			Disable	Enable							
8	54525953	MI1	13	1				1							3	Shut off	Full open	Auto	
9	58720257	MO1	14	1				0							3	Shut off	Full open	Auto	
10	79691777	MV1	19	1				1							3	Shut off	Full open	Auto	
11	25165825	CAL1_1	6	1															

	A	B	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI
1	#75	77	17	134	126	65-2	187	Event Enab	Event Enab	Event Enab	COV Enabl	COV Increı Event Det
22	96468993	AC1				1.E+09						

3. **Configuring State-Text property:**

Users can configure 10 state text values from CSV file.

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	110-1	BACnetARRAY of CharacterString	Multi-State Input (13) Multi-State Output (14) Multi-State Value (19)	0 to 255 characters
2	110-2			
3	110-3			
4	110-4			
5	110-5			
6	110-6			
7	110-7			
8	110-8			
9	110-9			
10	110-10			

	A	B	O	P	Q	R	S	T	U	V	W	X	Y
1	#75	77	74	110-1	110-2	110-3	110-4	110-5	110-6	110-7	110-8	110-9	110-10
8	54525953	MI1		Shut off	Full open	Auto							
9	58720257	MO1		Shut off	Full open	Auto							
10	79691777	MV1		Shut off	Full open	Auto							
11	25165825	CAL1_1											

4. Configuring Event-Enable property:

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Event Enable To Offnormal	BACnetEventTransitionBits	Analog Input (0) Analog Output (1) Analog Value (2) Binary Input (3) Binary Output (4) Binary Value (5) Multi-State Input (13) Multi-State Output (14) Notification Class (15) Schedule (17) Multi-State Value (19) Accumulator (23)	0 or 1
2	Event Enable To Fault			0 or 1
3	Event Enable To Normal			0 or 1

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	#75	77	79	-1	117	65-1	69	Event Enable To Offnormal	Event Enable To Fault	Event Enable To Normal	COV Enable	COV Increment	Event D
2	1	AI1	0	1	5	1000	0	1	1	1	1	9999999	
3	4194305	AO1	1	1	5	1000	0	1	1	1	1	9999999	
4	8388609	AV1	2	1	5	1000	0	1	1	1	1	9999999	
5	12582913	BI1	3	1				1	1	1	1		
6	16777217	BO1	4	1				1	1	1	1		
7	20971521	BV1	5	1				1	1	1	1		
8	54525953	MI1	13	1				1	1	1	1		
9	58720257	MO1	14	1				1	1	1	1		
10	79691777	MV1	19	1				1	1	1	1		

5. Configuring COV-Increment property:

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	COV Increment	Real	Analog Input (0) Analog Output (1) Analog Value (2)	-2,147,483,648 to 2,147,483,647

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	#75	77	79	-1	117	65-1	69	Event Enable To Offnorma	Event Enable To Fault	Event Enable To Normal	COV Enable	COV Increment	Event Dete	Alarm Valu
2	1	AI1	0	1	5	1000	0	1	1	1	1	2	1	
3	4194305	AO1	1	1	5	1000	0	1	1	1	1	5	1	
4	8388609	AV1	2	1	5	1000	0	1	1	1	1	9999999	1	
5	1.3E+07	BI1	3	1				1	1	1	1		1	1
6	1.7E+07	BO1	4	1				1	1	1	1		1	
7	2.1E+07	BV1	5	1				1	1	1	1		1	1
8	5.5E+07	MI1	13	1				1	1	1	1		1	
9	5.9E+07	MO1	14	1				1	1	1	1		1	
10	8E+07	MV1	19	1				1	1	1	1		1	

6. Configuring Event-Detection-Enable property:

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Event Detection Enable	Boolean	Analog Input (0) Analog Output (1) Analog Value (2) Binary Input (3) Binary Output (4) Binary Value (5) Multi-State Input (13) Multi-State Output (14) Notification Class (15) Schedule (17) Multi-State Value (19) Accumulator (23)	0 or 1

	A	B	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR
1	#75	77	COV Enabl	COV Incre	Event Detection Enable	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu
2	1	AI1			1								
3	4194305	AO1			1								
4	8388609	AV1			1								
5	12582913	BI1			1								
6	16777217	BO1			1								
7	20971521	BV1			1								
8	54525953	MI1			1								
9	58720257	MO1			1								
10	79691777	MV1			1								
16	71303169	SCH1			0								
21	62914561	NC1			0								
22	96468993	AC1			1								

7. Configuring Alarm-Value property:

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Alarm Value	BACnetBinaryPV	Binary Input (3) Binary Value (5)	0 or 1

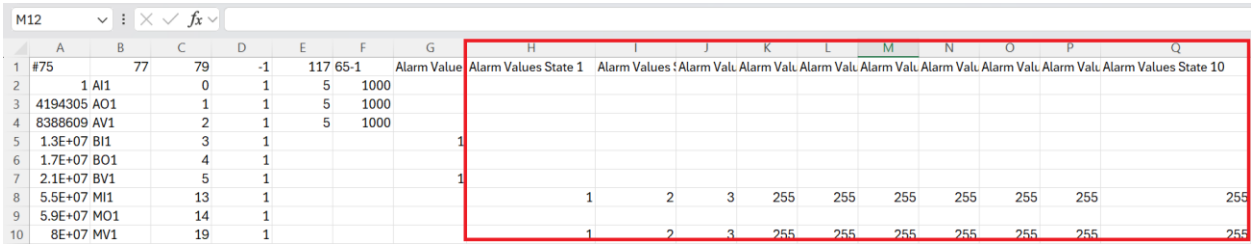
M12

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	#75	77	79	-1	117	65-1	Alarm Value	Alarm Values State 1	Alarm Values	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Values State 10
2	1	AI1	0	1	5	1000											
3	4194305	AO1	1	1	5	1000											
4	8388609	AV1	2	1	5	1000											
5	1.3E+07	BI1	3	1			1										
6	1.7E+07	BO1	4	1													
7	2.1E+07	BV1	5	1			1										
8	5.5E+07	MI1	13	1				1	2	3	255	255	255	255	255	255	255
9	5.9E+07	MO1	14	1													
10	8E+07	MV1	19	1				1	2	3	255	255	255	255	255	255	255

8. Configuring Alarm-Values property:

Users can configure 10 alarm values from CSV file.

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Alarm Values State 1	BACnetLIST of Unsigned	Multi-State Input (13) Multi-State Value (19)	0 to 255
2	Alarm Values State 2			
3	Alarm Values State 3			
4	Alarm Values State 4			
5	Alarm Values State 5			
6	Alarm Values State 6			
7	Alarm Values State 7			
8	Alarm Values State 8			
9	Alarm Values State 9			
10	Alarm Values State 10			



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	#75	77	79	-1	117	65-1	Alarm Value	Alarm Values State 1	Alarm Values	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Valu	Alarm Values State 10
2	1	AI1	0	1	5	1000											
3	4194305	AO1	1	1	5	1000											
4	8388609	AV1	2	1	5	1000											
5	1.3E+07	BI1	3	1			1										
6	1.7E+07	BO1	4	1													
7	2.1E+07	BV1	5	1			1										
8	5.5E+07	MI1	13	1				1	2	3	255	255	255	255	255	255	255
9	5.9E+07	MO1	14	1													
10	8E+07	MV1	19	1				1	2	3	255	255	255	255	255	255	255

9. Configuring Fault-Values property:

Users can configure 10 fault values from CSV file.

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Fault Values State 1	BACnetLIST of Unsigned	Multi-State Input (13) Multi-State Value (19)	0 to 255
2	Fault Values State 2			
3	Fault Values State 3			
4	Fault Values State 4			
5	Fault Values State 5			
6	Fault Values State 6			
7	Fault Values State 7			
8	Fault Values State 8			
9	Fault Values State 9			
10	Fault Values State 10			

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	#75	77	79	-1	117	65-1	69	Fault Values State 1	Fault Values State 2	Fault Valu	Fault Valu	Fault Valu	Fault Valu	Fault Valu	Fault Valu	Fault Valu	Fault Values State 10	Event Prio	Event
2	1	AI1	0	1	5	1000	0												
3	4194305	AO1	1	1	5	1000	0												
4	8388609	AV1	2	1	5	1000	0												
5	1.3E+07	BI1	3	1															
6	1.7E+07	BO1	4	1															
7	2.1E+07	BV1	5	1															
8	5.5E+07	MI1	13	1				255	255	255	255	255	255	255	255	255	255		
9	5.9E+07	MO1	14	1															
10	8E+07	MV1	19	1				255	255	255	255	255	255	255	255	255	255		

10. Configuring Priority property :

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Event Priority To Offnormal	BACnetARRAY[3] of Unsigned	Notification Class (15)	0 to 255
2	Event Priority To Fault			0 to 255
3	Event Priority To Normal			0 to 255

	A	B	BE	BF	BG	BH	BI	BJ
1	#75	77	lue Fault Valu	Event Priority To Offnormal	Event Priority To Normal	Event Priority To Fault	Event Reci	Event Reci
16	71303169	SCH1						
21	62914561	NC1		128	192	96		
22	96468993	AC1						

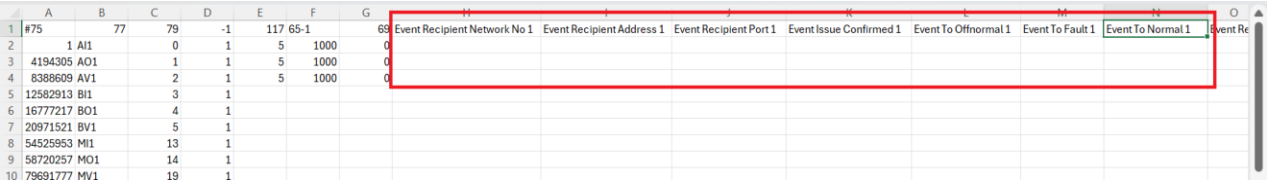
11. Configuring Recipient-List property:

Users can configure 10 recipient destinations from CSV file. Below are parameters of Recipient-List property,

a. Event Recipient Network No 1 to 10 : Network number of destination device.
b. Event Recipient Address 1 to 10 : IP Address of destination device.
c. Event Recipient Port 1 to 10 : Port number of destination device.
d. Event Issue Confirmed 1 to 10 : 1 / 0.
e. Event To Offnormal 1 to 10 : 1 / 0.
f. Event To Fault 1 to 10 : 1 / 0.
g. Event To Normal 1 to 10 : 1 / 0.

#	CSV Column Value	BACnet Datatype	Object Type Supported	Range Value
1	Event Recipient Network No 1	BACnetLIST of BACnetDestination	Notification Class (15)	0 to 65535
2	Event Recipient Address 1			Destination device IP Address (e,g, 172.16.1.100)
3	Event Recipient Port 1			0 to 65535 (e.g. 47808)
4	Event Issue Confirmed 1			0 : For Unconfirmed request or 1 : For Confirmed request

5	Event To Offnormal 1			0 or 1
6	Event To Fault 1			0 or 1
7	Event To Normal 1			0 or 1



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	#75	77	79	-1	117	65-1	69	Event Recipient Network No 1	Event Recipient Address 1	Event Recipient Port 1	Event Issue Confirmed 1	Event To Offnormal 1	Event To Fault 1	Event To Normal 1	Event Re
2	1	AI1	0	1	5	1000	0								
3	4194305	AO1	1	1	5	1000	0								
4	8388609	AV1	2	1	5	1000	0								
5	12582913	BI1	3	1											
6	16777217	BO1	4	1											
7	20971521	BV1	5	1											
8	54525953	MI1	13	1											
9	58720257	MO1	14	1											
10	79691777	MV1	19	1											

Note that BACnet/IP devices shall support at a minimum UDP port ranges 47808 – 47823 and 49152 – 65535 as per ANNEX J in ANSI/ASHRAE Standard 135-2016.